

ESP32-Based Indoor Positioning System Using RSSI Trilateration And Machine Learning

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ESP32-Based Indoor Positioning System Using RSSI Trilateration And Machine Learning

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Declaration

We hereby declare that this dissertation is the product of our own work, and we attest that it contains no material that resulted from collaboration, except where explicitly acknowledged in the text. Furthermore, we confirm that this project has not been previously submitted, either in part or in its entirety, to any other University or Institution for the purpose of obtaining any degree, diploma, or other qualification. All sources used and referenced in this dissertation are duly credited, and any borrowed ideas or information are appropriately cited in accordance with academic standards and guidelines.

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Certificate

This is to certify that the project entitled "ESP32-Based Indoor Positioning System Using RSSI Trilateration and Machine Learning", submitted by Debashnu Paul, Aryan Basu, Saswata Ghosh, Jishnu Baruah to Sister Nivedita University, West Bengal for the award of the degree of Bachelor of Technology (CSE) is a bonafide record of the project work carried out by them under my supervision and guidance. The content of the project, in full or parts have not been submitted to any other institute or university for the award of any degree or diploma.

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Abstract

Indoor positioning systems have gained significant importance in applications ranging from asset tracking to navigation in GPS-denied environments. This project presents the design and implementation of a low-cost, reproducible indoor positioning system using ESP32 microcontrollers, leveraging Received Signal Strength Indicator (RSSI) measurements and the ESP-NOW communication protocol. The system employs a three-phase architectural evolution that balances accuracy, scalability, and implementation complexity.

The first phase implements a hands-free single-target positioning system using trilateration with calibrated path-loss models, achieving expected precision of $\pm 5\text{--}10$ cm within a controlled 70×50 cm testbed. The second phase extends the system to support multiple targets through sequential time-slot broadcasting, maintaining precision of $\pm 10\text{--}15$ cm while preventing signal collisions. The third phase introduces a fully distributed mesh network architecture where all nodes measure pairwise RSSI values, enabling autonomous positioning with expected precision of ± 20 cm without centralized coordination.

A key contribution of this work is the systematic calibration protocol for deriving environment-specific path-loss constants (A and n) through controlled data collection. The project also explores machine learning approaches using Multi-Layer Perceptron (MLP) regression to directly map RSSI fingerprints to spatial coordinates, demonstrating the feasibility of neural network-based positioning as an alternative to geometric trilateration.

The system utilizes standardized data packet structures and synchronization mechanisms to ensure compatibility across all three phases, allowing incremental deployment and testing. Experimental validation is conducted using synthetic datasets and real-world RSSI measurements collected from the ESP32 testbed. Results demonstrate the trade-offs between positioning accuracy, system complexity, and scalability, providing insights for practical indoor localization deployments.

This project establishes a foundation for future enhancements including dynamic environment adaptation, integration with existing infrastructure, and advanced algorithms such as graph optimization and Simultaneous Localization and Mapping (SLAM) for large-scale networks.

Chapter 1

Introduction

1.1 Project Overview and Rationale

Indoor positioning systems have become increasingly critical in modern applications where Global Positioning System (GPS) signals are unavailable or unreliable. From warehouse asset tracking and hospital patient monitoring to augmented reality navigation and smart building automation, the demand for accurate, cost-effective indoor localization solutions continues to grow. While GPS provides excellent outdoor positioning accuracy, its signals cannot penetrate building structures effectively, creating a significant gap in location-based services for indoor environments [1, 2].

This project presents the design and implementation of an ESP32-based Indoor Positioning System (IPS) that leverages Received Signal Strength Indicator (RSSI) measurements and the ESP-NOW communication protocol to achieve low-cost, reproducible indoor localization. The ESP32 microcontroller, with its integrated Wi-Fi capabilities, dual-core processor, and low power consumption, provides an ideal platform for developing scalable wireless sensor networks [3]. ESP-NOW, a connectionless communication protocol developed by Espressif Systems, enables fast, low-latency data transmission between devices without the overhead of traditional Wi-Fi association, making it particularly suitable for real-time positioning applications [4].

The fundamental principle underlying this system is RSSI-based distance estimation using path-loss models. As radio frequency signals propagate through space, their strength diminishes predictably with distance, following the relationship: $d = 10^{(A-RSSI)/(10n)}$, where d is the distance, RSSI is the received signal strength, A is the signal strength at 1 meter reference distance, and n is the path-loss exponent dependent on the environment [5, 6]. By measuring RSSI values from multiple fixed anchor nodes with known positions, the location of a target node can be estimated through trilateration or machine learning regression techniques [7, 8].

1.1.1 Low-Cost Reproducible Testbed Approach

A distinguishing feature of this work is its emphasis on reproducibility and accessibility through a low-cost hardware platform. The system is designed with reproducibility and scalability as core principles. All nodes remain stationary during measurement periods to eliminate motion-induced interference, and a standardized 70×50 cm testbed with precisely positioned anchors

ensures consistent experimental conditions. The use of readily available ESP32 microcontrollers, simple grid-based layout, and open-source software enables replication and extension by other researchers and practitioners with minimal investment. This approach democratizes indoor positioning research by removing barriers associated with expensive proprietary hardware and complex infrastructure requirements.

1.1.2 Three-Phase Architectural Evolution

A key innovation of this work is the three-phase architectural evolution that balances accuracy, scalability, and implementation complexity. The first phase implements a *Hands-Free (Single Target)* system where a stationary target node broadcasts packets to fixed anchor nodes, enabling calibration of environment-specific path-loss constants and validation of trilateration algorithms with expected precision of $\pm 5\text{--}10$ cm. The second phase extends to *Multi-Target (Sequential Broadcasts)* operation using time-slot coordination, allowing multiple target nodes to coexist without signal collision while maintaining $\pm 10\text{--}15$ cm precision. The third phase introduces an *All-Nodes/Mesh Mode* where every node acts as both transmitter and receiver, creating a fully distributed network capable of autonomous positioning with ± 20 cm relative precision through pairwise RSSI measurements and graph optimization algorithms [9, 10].

The use of standardized data packet structures and synchronization mechanisms ensures compatibility across all three phases, allowing incremental deployment and testing. This phased approach not only simplifies development and debugging but also provides clear migration paths for different application requirements. Each phase builds upon the previous one, maintaining compatibility with ESP-NOW data structures and protocols, ensuring that lessons learned and calibration data from earlier phases remain valuable as the system evolves.

Beyond traditional geometric positioning methods, this project explores machine learning approaches using Multi-Layer Perceptron (MLP) regression to directly map RSSI fingerprints to spatial coordinates. By training neural networks on collected RSSI datasets, the system can potentially achieve better accuracy in complex environments where multipath fading and environmental variations make analytical models less reliable [11, 12]. The project includes analysis of an existing MLPRegressor model trained on synthetic data and evaluates the feasibility of improving performance with real experimental measurements.

This work contributes to the growing body of research on low-cost indoor positioning by demonstrating a practical, reproducible system that can be deployed with minimal hardware investment. The phased architecture provides flexibility for different use cases, from simple single-target tracking to complex mesh network deployments. The systematic calibration protocol and standardized data structures facilitate replication and extension of this work for various indoor environments and application domains.

1.2 Objectives

The primary objectives of this project are:

1. **Model Derivation and Calibration:** Establish a systematic calibration protocol to derive

environment-specific path-loss constants (A and n) through controlled RSSI measurements at known distances. Validate the path-loss model: $d = 10^{(A-\text{RSSI})/(10n)}$ for accurate distance estimation in the testbed environment.

2. **Protocol Development:** Design and implement a sequential time-slot broadcasting protocol that enables multiple target nodes to transmit without signal collision. Develop standardized data packet structures (`target_packet_t` and `anchor_data_t`) that maintain compatibility across all system phases.
3. **Network Implementation:** Implement three progressive system architectures: (a) Hands-Free single-target mode for calibration and baseline accuracy, (b) Multi-target mode with sequential timing for scalability, and (c) All-Nodes mesh mode for fully distributed autonomous positioning.
4. **Accuracy Validation:** Achieve and validate positioning precision targets of $\pm 5\text{--}10$ cm for single-target mode, $\pm 10\text{--}15$ cm for multi-target mode, and ± 20 cm for mesh network mode through systematic experimental testing on the 70×50 cm testbed.
5. **Machine Learning Integration:** Analyze the performance of MLPRegressor neural network for direct RSSI-to-coordinate mapping, evaluate its feasibility as an alternative to geometric trilateration, and identify strategies for improving model accuracy with real experimental data.

1.3 Contributions

This project makes several key contributions to indoor positioning system research and implementation:

1. **Phased Compatible Architecture:** A three-phase system evolution (Hands-Free $\rightarrow$ Multi-Target $\rightarrow$ Mesh) that maintains backward compatibility through standardized protocols and data structures. This approach allows incremental deployment, simplified debugging, and clear migration paths for different application requirements.
2. **Standardized Data Packet Structures:** Well-defined `target_packet_t` and `anchor_data_t` structures that encapsulate node identification, RSSI measurements, and timestamps in a format compatible with ESP-NOW protocol constraints. These structures facilitate data aggregation, processing, and analysis across all system phases.
3. **Systematic Calibration Protocol:** A reproducible methodology for deriving environment-specific path-loss parameters through controlled measurements at known grid positions. The protocol includes guidelines for sample collection (50–100 samples per anchor), RSSI filtering techniques, and validation procedures to ensure model accuracy.

4. **Synchronization Mechanisms:** Time-slot coordination algorithms that prevent packet collisions in multi-target scenarios while maintaining low latency. The synchronization approach uses unique node IDs, configurable transmission intervals (100ms spacing), and optional gateway coordination for scalable network operation.
5. **Hybrid Positioning Approach:** Integration of both geometric (trilateration) and machine learning (MLP regression) methods, providing comparative analysis of their accuracy, computational requirements, and suitability for different deployment scenarios. This dual approach offers flexibility in choosing positioning algorithms based on application constraints.
6. **Low-Cost Reproducible Testbed:** A practical implementation using readily available ESP32 microcontrollers, simple grid-based layout, and open-source software that enables replication and extension by other researchers and practitioners with minimal investment.

1.4 Organization of the Chapters

The remainder of this report is organized as follows:

Chapter 2: Literature Review surveys existing research on RSSI-based indoor positioning, covering fundamental concepts of path-loss modeling, localization techniques (trilateration, multilateration, machine learning), ESP-NOW protocol characteristics, distributed network architectures, and relevant positioning algorithms. This chapter establishes the theoretical foundation for the proposed system.

Chapter 3: Problem Identification articulates the key challenges in RSSI-based indoor positioning, including signal instability due to multipath fading, environmental dependency of propagation parameters, motion interference, and scalability issues in multi-target scenarios. The chapter connects these challenges to the phased architectural approach adopted in this project.

Chapter 4: Proposed Methodology presents the detailed system design, including the three-phase architecture, hardware setup specifications, data communication flows, packet structure definitions, and synchronization mechanisms. This chapter includes system architecture diagrams, data flow visualizations, and hardware layout schematics that illustrate the implementation approach.

Chapter 5: Experimental Results and Discussion documents the calibration protocol execution, dataset structure and collection methodology, machine learning model analysis using synthetic data, feasibility assessment for model improvement with real measurements, data processing pipeline, and expected positioning accuracy for each system phase. Performance metrics and comparative analysis are presented.

Chapter 6: Conclusion and Future Work summarizes the project achievements, highlights key takeaways regarding calibration importance and sequential timing benefits, outlines the evolution roadmap for mesh network implementation, and discusses potential improvements

including dynamic environment adaptation, integration with existing infrastructure, and advanced algorithms for large-scale deployments.

Chapter 2

Literature Survey

This chapter reviews the theoretical and practical foundations relevant to the proposed ESP32 Indoor Positioning System. The review focuses on RSSI-based localization fundamentals, localization techniques, low-power wireless protocols, distributed network architectures, and machine learning approaches for indoor positioning.

2.1 RSSI-Based Positioning Fundamentals

Received Signal Strength Indicator (RSSI) is one of the most widely adopted metrics for low-cost proximity and distance estimation in wireless sensor networks [1]. Its simplicity and availability in most wireless communication protocols make it an attractive choice for indoor positioning systems. However, RSSI measurements are susceptible to environmental noise, multipath fading, and shadowing effects, which introduce significant challenges in achieving high positioning accuracy [2].

2.1.1 Path-Loss Modeling

The relationship between distance and signal strength is commonly modeled using the log-distance path loss model, which provides a mathematical framework for converting RSSI measurements into distance estimates [5]. The core equation is given by:

$$d = 10^{\left(\frac{A - \text{RSSI}}{10n}\right)} \quad (2.1)$$

where d is the estimated distance in meters, A is the RSSI value measured at a reference distance (typically 1 meter), RSSI is the measured signal strength in dBm, and n is the path-loss exponent that characterizes the rate at which signal strength decreases with distance.

The parameters A and n are highly environment-dependent and vary based on factors such as room layout, furniture, wall materials, and the presence of people or moving objects [13]. Literature confirms that both parameters must be derived through careful, localized calibration to mitigate estimation errors and achieve acceptable positioning accuracy [6, 14].

2.1.2 Calibration Requirements

Calibration is a critical step in RSSI-based positioning systems. Without proper calibration, the path-loss model parameters may not accurately reflect the actual propagation characteristics of the deployment environment, leading to significant distance estimation errors [15]. The calibration process typically involves:

- Collecting RSSI samples at known distances in the target environment
- Performing regression analysis to determine optimal values for A and n
- Validating the calibrated model across different locations within the environment
- Periodically recalibrating to account for environmental changes

Studies have shown that proper calibration can reduce distance estimation errors by 30-50% compared to using generic path-loss parameters [16].

2.2 Localization Techniques

Various algorithms have been developed to convert distance estimates or RSSI measurements into position coordinates. The choice of technique depends on the system architecture, number of reference nodes, and computational resources available.

2.2.1 Trilateration

Trilateration is the most fundamental localization technique, requiring distance measurements from at least three fixed anchor nodes with known positions [2]. Given the distances d_1, d_2, d_3 from three anchors at positions (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , the target position (x, y) can be determined by solving the system of equations:

$$(x - x_i)^2 + (y - y_i)^2 = d_i^2, \quad i = 1, 2, 3 \quad (2.2)$$

In practice, measurement noise and errors require the use of least-squares optimization or other robust estimation techniques [17]. Trilateration is particularly suitable for single-target scenarios where the system architecture follows a star topology with fixed anchors and a mobile target.

2.2.2 Multilateration and Multidimensional Scaling

For mesh network configurations where multiple nodes measure pairwise distances, multilateration and Multidimensional Scaling (MDS) techniques become more appropriate [18]. MDS algorithms reconstruct the relative topology of the entire network by minimizing the stress

function that measures the discrepancy between measured distances and Euclidean distances in the reconstructed coordinate system [19].

These techniques are particularly valuable in distributed positioning systems where no fixed infrastructure exists, and all nodes cooperatively determine their relative positions. The approach is robust to individual measurement errors as it leverages redundant distance measurements across the network [10].

2.2.3 Machine Learning Regression

Recent advances in machine learning have introduced data-driven approaches that bypass explicit distance conversion and directly map RSSI measurements to position coordinates [8]. Neural network architectures, particularly Multi-Layer Perceptrons (MLPs) and Recurrent Neural Networks (RNNs), can learn the complex non-linear relationship between RSSI fingerprints and spatial locations [11].

The key advantage of ML-based approaches is their ability to implicitly model environmental effects, multipath propagation, and other factors that are difficult to capture in analytical models [20]. However, these methods require substantial training data collected in the target environment and may not generalize well to different settings without retraining [12].

2.3 Low-Power Wireless Protocols

The choice of wireless communication protocol significantly impacts the performance, power consumption, and scalability of indoor positioning systems.

2.3.1 ESP-NOW Protocol

ESP-NOW is a connectionless communication protocol developed by Espressif Systems, built on the IEEE 802.11 physical layer [4]. Unlike traditional Wi-Fi, ESP-NOW does not require association with an access point or the overhead of maintaining TCP/IP connections. This makes it particularly suitable for applications requiring fast, low-latency transmission of small data packets [3].

Key advantages of ESP-NOW for indoor positioning include:

- **Low Latency:** Packet transmission occurs within milliseconds, enabling rapid sequential measurements
- **Minimal Overhead:** No connection establishment or handshaking required
- **Broadcast Support:** One-to-many communication simplifies data collection from multiple anchors
- **Low Power Consumption:** Reduced protocol overhead translates to lower energy consumption

2.3.2 Comparison with Wi-Fi and Bluetooth

Traditional Wi-Fi (IEEE 802.11) provides higher data rates but requires association with an access point and involves significant protocol overhead for connection management [21]. This makes it less suitable for time-sensitive, sequential RSSI measurements where rapid data collection is critical.

Bluetooth Low Energy (BLE) offers good power efficiency and is widely used in indoor positioning systems [22]. However, BLE's connection-oriented nature and lower data rates compared to ESP-NOW can introduce latency in multi-node scenarios. Additionally, BLE's advertising mechanism may not provide the same level of control over transmission timing as ESP-NOW's broadcast mode.

For the specific requirements of this project-rapid sequential broadcasts, minimal latency, and precise timing control-ESP-NOW provides an optimal balance of performance and simplicity [23].

2.4 Distributed Network Architectures

As positioning systems evolve from centralized star topologies to distributed mesh networks, new architectural considerations and algorithms become necessary.

2.4.1 Adjacency Map Construction

In a mesh network configuration, every node can measure the RSSI to every other node within communication range, creating a comprehensive adjacency map [9]. This map contains entries of the form $(Node_i, Node_j, RSSI_{i \rightarrow j})$, representing the signal strength observed by node i when receiving transmissions from node j .

The adjacency map serves as the foundation for distributed positioning algorithms, enabling each node to estimate its position relative to its neighbors without requiring a centralized coordinator [24]. This approach enhances system robustness by eliminating single points of failure and allows for dynamic network reconfiguration as nodes join or leave the network.

2.4.2 Graph Optimization Approaches

Advanced distributed positioning systems treat the network as a graph where nodes are vertices and pairwise distance estimates are edges [10]. Graph optimization algorithms iteratively refine the global position estimates by minimizing a cost function that measures the inconsistency between measured distances and Euclidean distances in the estimated coordinate system.

Semidefinite programming (SDP) relaxation techniques have been successfully applied to solve the network localization problem, providing convex optimization formulations that can be solved efficiently even for large networks [25]. These approaches are particularly robust

to measurement noise and can handle incomplete distance information where not all pairwise distances are available.

2.4.3 SLAM-Style Positioning

Techniques borrowed from Simultaneous Localization and Mapping (SLAM) in robotics can be adapted for distributed indoor positioning [10]. In this paradigm, nodes incrementally build a map of the network topology while simultaneously estimating their own positions. The approach uses probabilistic frameworks such as Extended Kalman Filters (EKF) or particle filters to maintain uncertainty estimates and update positions as new measurements become available.

SLAM-style approaches are particularly valuable in dynamic environments where nodes may be mobile or where the network topology changes over time. The probabilistic framework naturally handles measurement uncertainty and provides confidence bounds on position estimates.

2.5 Machine Learning Approaches for Indoor Positioning

Machine learning has emerged as a powerful tool for indoor positioning, offering alternatives to traditional geometric approaches.

2.5.1 Neural Networks for Indoor Positioning

Neural networks can learn complex mappings from RSSI measurements to position coordinates without requiring explicit modeling of signal propagation [8]. Multi-Layer Perceptrons (MLPs) with multiple hidden layers have been successfully applied to indoor positioning, achieving accuracy comparable to or better than traditional trilateration methods [12].

Deep learning architectures, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have shown promise in capturing spatial and temporal patterns in RSSI data [11]. RNNs, in particular, can model the temporal dynamics of RSSI measurements, potentially improving robustness to short-term fluctuations and noise.

2.5.2 MLPRegressor Architecture

The Multi-Layer Perceptron Regressor (MLPRegressor) is a supervised learning algorithm that uses backpropagation to learn the relationship between input features (RSSI values from multiple anchors) and output targets (X, Y coordinates) [20]. The architecture typically consists of:

- **Input Layer:** RSSI measurements from all anchor nodes
- **Hidden Layers:** Multiple layers of neurons with non-linear activation functions (e.g., ReLU)

- **Output Layer:** Two neurons representing X and Y coordinates

The network is trained using gradient descent optimization to minimize the mean squared error between predicted and actual positions. Regularization techniques such as dropout or L2 regularization can be applied to prevent overfitting, especially when training data is limited.

2.5.3 Feature Engineering from RSSI Data

The performance of machine learning models can be significantly enhanced through careful feature engineering [26]. Beyond raw RSSI values, derived features can provide additional information:

- **Distance Ratios:** Ratios of distances to different anchors can be more stable than absolute distances
- **Signal Gradients:** Temporal derivatives of RSSI measurements can capture motion patterns
- **Statistical Features:** Mean, variance, and other statistics computed over sliding windows
- **Spatial Features:** Differences or ratios between RSSI values from different anchors

Feature normalization using techniques such as StandardScaler is essential to ensure that all features contribute equally to the learning process and to improve convergence during training [27].

2.5.4 Comparison with Traditional Methods

Machine learning approaches offer several advantages over traditional geometric methods:

- Implicit modeling of environmental effects and multipath propagation
- No need for explicit calibration of path-loss parameters
- Ability to learn from data and adapt to specific environments
- Potential for higher accuracy in complex, non-line-of-sight scenarios

However, ML methods also have limitations:

- Require substantial training data collected in the target environment
- May not generalize well to different environments without retraining
- Computational requirements may be higher, especially for deep architectures
- Lack of interpretability compared to physics-based models

The choice between ML and traditional methods depends on the specific application requirements, available computational resources, and the feasibility of collecting sufficient training data [28].

Chapter 3

Problem Identification

Achieving high-precision indoor localization with a low-cost system presents unique challenges that must be systematically addressed by the methodology.

3.1 RSSI Instability and Environmental Noise

The primary technical hurdle is the inherent instability of the RSSI metric in typical indoor environments[29]. Unlike outdoor GPS systems that benefit from line-of-sight satellite signals, indoor positioning systems must contend with complex signal propagation patterns that introduce significant measurement uncertainty.

3.1.1 Multipath Fading Effects

Indoor spaces are highly reflective environments where radio signals bounce off walls, ceilings, furniture, and other obstacles. This phenomenon, known as multipath fading, causes the receiver to detect multiple copies of the transmitted signal arriving at different times and with varying phase relationships[30]. The constructive and destructive interference between these signal paths leads to rapid, unpredictable fluctuations in RSSI measurements, which severely corrupts distance calculations[31]. Even in a static environment, RSSI values can vary by 5-10 dBm over short time intervals, translating to significant positioning errors when using path-loss models.

3.1.2 Environmental Dependency of Path-Loss Parameters

The path-loss model used for distance estimation, expressed as $d = 10^{(A-RSSI)/(10n)}$, relies on two critical parameters: the reference signal strength at 1 meter (A) and the path-loss exponent (n)[32]. These parameters are highly specific to the physical environment, including room layout, wall materials, furniture density, and even the presence of people[33]. A value of n that works in an open laboratory space may be completely inappropriate for a cluttered office environment. Failure to calibrate and derive these constants locally for each deployment environment will result in unacceptable error margins, potentially exceeding several meters even in small indoor spaces.

3.1.3 Motion Interference

Even minor movements of objects or people within the measurement area can drastically alter RSSI readings[34]. Human bodies, being primarily composed of water, are particularly effective at absorbing and reflecting 2.4 GHz signals. A person walking between a transmitter and receiver can cause RSSI drops of 10-20 dBm, temporarily invalidating distance estimates[35]. Additionally, electrical interference from power supplies and other electronic devices can introduce noise into the measurements. The proposed solution addresses these challenges by mandating that all ESP32 nodes remain completely static during sampling periods and by using battery power to eliminate electrical noise from AC adapters, ensuring clean and reproducible signal measurements.

3.2 The Challenge of System Scalability

While single-target positioning systems can achieve reasonable accuracy through careful calibration, scaling the IPS from a single target to a multi-node environment introduces major protocol-level challenges[36]. The transition from one transmitter to multiple concurrent transmitters fundamentally changes the communication dynamics and requires sophisticated coordination mechanisms to maintain system reliability.

3.2.1 Signal Collision in Multi-Target Scenarios

When multiple target nodes transmit simultaneously using the ESP-NOW broadcast protocol, their radio signals can collide in the shared 2.4 GHz spectrum[37]. Unlike connection-oriented protocols that implement carrier sense multiple access (CSMA), ESP-NOW's connectionless nature provides no inherent collision avoidance mechanism[38]. When two or more nodes transmit packets at overlapping times, the resulting interference causes packet corruption at the receiver. This leads to complete packet loss, where anchors fail to receive any usable data, or partial corruption, where the packet is received but contains invalid RSSI measurements. In a system with N targets transmitting at random intervals, the collision probability increases quadratically, making uncoordinated multi-target operation impractical for reliable positioning.

3.2.2 Ambiguous RSSI Source Identification

Even when collision avoidance mechanisms partially succeed, overlapping transmissions create a fundamental ambiguity problem[39]. If an anchor receives signals from multiple targets within a short time window, it cannot reliably determine which target generated which RSSI reading. The ESP-NOW protocol includes a source MAC address in each packet, but signal overlap can corrupt this identifier. Without accurate source identification, the anchor cannot properly tag RSSI measurements with the correct target ID, resulting in ambiguous data that cannot be used for position estimation. This problem is particularly severe in dense deployments where

multiple targets may be equidistant from an anchor, producing similar RSSI values that become indistinguishable when corrupted.

3.2.3 RF Noise Floor Elevation

High concurrent transmission traffic in a multi-node system raises the overall radio frequency (RF) noise floor in the environment[40]. Each active transmitter contributes to the background electromagnetic energy, reducing the signal-to-noise ratio (SNR) for all nodes. As the noise floor rises, the effective communication range decreases, and RSSI measurements become less stable[41]. Weak signals that would normally be detectable in a quiet RF environment become buried in noise, leading to increased packet loss rates and reduced positioning accuracy. This effect is compounded in indoor environments where reflections and multipath propagation already create a challenging signal environment. The cumulative impact of multiple transmitters can reduce the effective positioning area by 30-50% compared to single-target operation.

3.3 Mitigation Strategy: Phased Architecture

To systematically overcome the identified challenges of RSSI instability and system scalability, this project employs a conservative, phased architectural approach[42]. Rather than attempting to solve all problems simultaneously, each phase addresses specific technical challenges while building upon the foundation established in previous phases. This methodology ensures that system accuracy and reliability are thoroughly validated before introducing additional complexity.

3.3.1 Phase 1: Calibration and Environmental Modeling

The first phase focuses exclusively on establishing a reliable baseline by eliminating all sources of variability[43]. In this single-target configuration, only one ESP32 node acts as a transmitter while all others function as fixed anchor receivers. By removing motion, multiple transmitters, and other dynamic factors, this phase isolates and characterizes the fundamental environmental noise and multipath effects specific to the deployment location. The primary objective is to derive accurate path-loss parameters (A and n) through systematic measurement at known distances[44]. This calibration data forms the foundation for all subsequent distance estimation, ensuring that the mathematical model accurately reflects the actual signal propagation characteristics of the environment. The static nature of this phase also allows for extensive sampling (50-100 measurements per position) to establish statistical confidence in the RSSI-to-distance relationship.

3.3.2 Phase 2: Sequential Timing for Multi-Target Operation

Once the environmental model is validated, the second phase introduces multiple target nodes while maintaining system reliability through time-division multiple access (TDMA)[45]. Each target is assigned a dedicated time slot during which it has exclusive access to the communication channel. For example, Target A transmits during the 0-100ms window, Target B during 100-200ms, and Target C during 200-300ms. This sequential approach completely eliminates the collision and ambiguity problems identified in Section 3.2[46]. Anchors continuously listen and can unambiguously associate each received packet with its source based on the transmission timing. The time-slot protocol also enables precise synchronization across the network, as all nodes share a common time reference. By spacing transmissions with sufficient guard intervals (typically 100ms), the system ensures that even delayed packets or retransmissions do not overlap with subsequent time slots. This phase demonstrates that the system can scale to multiple targets without sacrificing the positioning accuracy established in Phase 1.

3.3.3 Phase 3: Distributed Mesh Network Architecture

The final phase transforms the system into a fully distributed mesh network where every ESP32 node acts as both transmitter and receiver[47]. Building upon the sequential timing mechanism from Phase 2, each node is assigned a unique ID and time slot for transmission. During its allocated slot, a node broadcasts its identification packet, and all other nodes measure and record the RSSI value. This creates a comprehensive pairwise RSSI matrix that captures the signal strength between every pair of nodes in the network. The distributed nature of this architecture eliminates the need for a centralized gateway during measurement, as each node maintains its own local RSSI table. A gateway or PC can later aggregate these tables for network-wide position estimation using advanced algorithms such as multidimensional scaling (MDS), graph optimization, or SLAM-style techniques[48]. This phase addresses the scalability challenge by distributing both the measurement and computational load across all nodes, enabling the system to grow to larger networks without overwhelming any single component.

3.3.4 Problem-Solution Mapping

This phased evolution directly addresses each identified challenge:

- **RSSI Instability:** Phase 1's static calibration and extensive sampling establish reliable baseline measurements despite multipath fading and environmental noise.
- **Environmental Dependency:** Phase 1's local calibration derives environment-specific path-loss parameters, ensuring accurate distance estimation for the deployment location.
- **Motion Interference:** All phases mandate static node placement during measurement, eliminating dynamic interference sources.

- **Signal Collision:** Phase 2's time-slot protocol provides exclusive channel access, completely preventing packet collisions.
- **Source Ambiguity:** Sequential timing and unique node IDs enable unambiguous identification of RSSI sources.
- **RF Noise Floor:** Time-division reduces instantaneous transmitter count to one, maintaining a low noise floor throughout operation.

This structured evolution ensures that the system's accuracy is thoroughly established before scaling complexity, providing a robust foundation for reliable indoor positioning.

Chapter 4

Proposed Methodology

This chapter presents the comprehensive methodology for the ESP32 Indoor Positioning System, detailing the three-phase architectural evolution, hardware configuration, data communication protocols, and synchronization mechanisms. The proposed solution emphasizes a systematic progression from single-target calibration to multi-target sequential operation, culminating in a fully distributed mesh network [1, 9].

4.1 System Architectural Phases

The ESP32 IPS employs a phased architectural approach that ensures compatibility and incremental complexity management. Each phase builds upon the previous one while maintaining the core ESP-NOW communication protocol and data structures.

4.1.1 Phase 1: Hands-Free (Single Target)

Phase 1 establishes the foundational calibration and modeling framework. In this configuration, a single target node broadcasts ESP-NOW packets [4] while remaining stationary at known grid coordinates. Four anchor nodes, positioned at fixed corners of the testbed, receive these broadcasts and measure the Received Signal Strength Indicator (RSSI) values [6, 22].

The communication flow is strictly unidirectional: the target transmits periodic broadcast packets containing its identifier and timestamp, while anchors passively listen and record RSSI measurements. Each anchor then forwards its data to a gateway node, which aggregates the readings and transmits the complete dataset to a PC for logging and analysis.

This phase serves multiple critical purposes:

- **Calibration:** Deriving environment-specific path-loss constants (A, n) by collecting RSSI measurements at known distances
- **Model Validation:** Verifying the accuracy of the path-loss model [5, 13]: $d = 10^{(A-RSSI)/(10n)}$
- **Baseline Establishment:** Creating a reference dataset for machine learning model training

- **Protocol Testing:** Validating ESP-NOW communication reliability and RSSI measurement consistency

The static nature of all nodes during measurement periods eliminates motion-induced interference and ensures high reproducibility of results. Expected positioning precision in this phase is $\pm 5\text{-}10$ cm on the scaled testbed.

4.1.2 Phase 2: Multi-Target (Sequential Broadcasts)

Phase 2 extends the system to support multiple target nodes through time-slotted sequential transmission. This approach addresses the fundamental challenge of signal collision that arises when multiple nodes transmit simultaneously in the same frequency band.

The sequential protocol assigns each target node a dedicated time slot within a repeating cycle. For example, with three target nodes:

- Target A transmits during 0-100 ms
- Target B transmits during 100-200 ms
- Target C transmits during 200-300 ms
- Cycle repeats every 300 ms

Anchor nodes listen continuously across all time slots, recording both the RSSI value and the target identifier from each received packet. This tagged data enables the gateway to distinguish measurements from different targets and maintain separate position estimates for each.

Key advantages of sequential transmission include:

- **Collision Avoidance:** Guaranteed non-overlapping transmissions prevent packet loss
- **Source Disambiguation:** Clear target identification eliminates RSSI ambiguity
- **Scalability:** Additional targets can be accommodated by extending the cycle period
- **Noise Reduction:** Lower RF noise floor compared to simultaneous transmission

The time-slot protocol requires loose synchronization among nodes, achievable through the ESP32's internal `millis()` timer or an external Real-Time Clock (RTC). Expected positioning precision in this phase is $\pm 10\text{-}15$ cm per target.

4.1.3 Phase 3: All-Nodes Mesh Mode

Phase 3 represents the most advanced configuration, transforming the system into a fully distributed mesh network where every node functions as both transmitter and receiver. This bidirectional communication enables the construction of a complete RSSI adjacency map capturing pairwise signal strength measurements between all node pairs.

In mesh mode, each node periodically broadcasts its identifier while simultaneously listening for broadcasts from all other nodes. The resulting dataset contains entries of the form (Node_i, Node_j, RSSI(i→j), RSSI(j→i), Timestamp), providing bidirectional RSSI measurements for every node pair.

The gateway aggregates these pairwise measurements and applies advanced positioning algorithms:

- **Multidimensional Scaling (MDS):** Derives relative node positions from distance estimates [19, 18]
- **Graph Optimization:** Minimizes positioning error across the entire network [10, 25]
- **SLAM-style Approaches:** Simultaneously estimates node positions and refines the RSSI-distance model

Mesh mode offers several distinctive capabilities:

- **Distributed Autonomy:** No fixed anchor infrastructure required
- **Relative Positioning:** Nodes can determine their positions relative to each other
- **Network Resilience:** System continues functioning even if individual nodes fail
- **Scalability:** Network can expand dynamically as new nodes join

Expected positioning precision in mesh mode is ± 20 cm relative to neighboring nodes, with accuracy dependent on network density and environmental interference.

4.1.4 Phase Compatibility and Evolution

A critical design principle is maintaining compatibility across all three phases. The same ESP-NOW protocol, packet structures, and RSSI measurement techniques are used throughout, enabling seamless transitions between operational modes. Nodes can be reconfigured from anchor to target roles through software updates without hardware modifications.

This evolutionary approach provides several benefits:

- **Incremental Development:** Each phase can be developed and tested independently
- **Risk Mitigation:** Issues can be identified and resolved at each stage before adding complexity
- **Flexible Deployment:** System can operate in the mode most appropriate for specific applications
- **Educational Value:** Progressive complexity facilitates understanding of positioning principles

Figure 4.1 illustrates the three-phase architectural evolution and the communication patterns in each configuration.

4.2 Hardware Setup and System Layout

The ESP32 IPS testbed employs commercially available components configured in a controlled indoor environment to ensure measurement reproducibility and system reliability.

4.2.1 Component Specification

The system comprises the following hardware components:

- **ESP32 Microcontrollers:** 6-8 ESP32 development boards [3] (e.g., ESP32-WROOM-32) serving as anchor nodes, target nodes, and gateway. Each board features built-in Wi-Fi/Bluetooth capabilities, sufficient GPIO pins for peripheral connections, and support for the ESP-NOW protocol [4].
- **Power Supply:** Individual battery packs (e.g., 3.7V Li-ion or 5V USB power banks) for each node to eliminate ground loop interference and enable wireless operation. Battery power ensures cleaner RSSI measurements compared to mains-powered configurations.
- **OLED Displays:** Small OLED screens (e.g., 0.96" I2C OLED) connected to each node for real-time debugging, displaying node ID, RSSI values, packet counts, and operational status without requiring serial connection to a PC.
- **Testbed Surface:** Chart paper or grid mat measuring 70×50 cm with 10 cm grid spacing marked for precise target placement. The surface should be flat, non-metallic, and positioned away from reflective surfaces to minimize multipath effects.
- **Node Mounts:** Adjustable stands or foam blocks to maintain consistent antenna height (10 cm above the testbed surface) across all nodes. Height consistency is critical for RSSI measurement reproducibility.
- **Measuring Tools:** Precision measuring tape or ruler for accurate node placement at grid coordinates, ensuring positioning errors remain below ± 1 cm.

4.2.2 Testbed Layout and Coordinate System

The testbed employs a Cartesian coordinate system with the origin at the bottom-left corner. The coordinate space spans 70 cm in the X-axis and 50 cm in the Y-axis, providing a total area of 3,500 cm² for positioning experiments.

Four anchor nodes are positioned at the corners of the testbed:

- **Anchor 1 (A1):** Coordinates (0, 0) - Bottom-left corner
- **Anchor 2 (A2):** Coordinates (70, 0) - Bottom-right corner
- **Anchor 3 (A3):** Coordinates (0, 50) - Top-left corner

- **Anchor 4 (A4):** Coordinates (70, 50) - Top-right corner

This corner placement maximizes the geometric diversity of RSSI measurements, providing optimal conditions for trilateration and multilateration algorithms [17, 24]. The rectangular configuration ensures that target positions within the testbed are surrounded by anchors, improving positioning accuracy [14].

Target nodes are placed at grid intersections during calibration, typically at 10 cm intervals: (10,10), (20,10), (30,10), ..., (60,40). This systematic sampling provides comprehensive coverage of the testbed area and enables accurate path-loss model derivation.

Figure 4.3 illustrates the complete testbed configuration with anchor positions, coordinate system, and sample target placements.

4.2.3 Antenna Height and Environmental Considerations

All ESP32 nodes are mounted at a consistent antenna height of approximately 10 cm above the testbed surface. This elevation serves multiple purposes:

- **Ground Reflection Mitigation:** Reduces multipath interference from surface reflections
- **Measurement Consistency:** Ensures comparable RSSI values across different node positions
- **Practical Relevance:** Approximates real-world scenarios where devices are held or placed at table height

Environmental factors are carefully controlled during experiments:

- **Static Configuration:** All nodes remain completely stationary during measurement periods to eliminate motion-induced RSSI fluctuations
- **Interference Avoidance:** Testbed is positioned away from metal objects, active Wi-Fi access points, and other 2.4 GHz sources
- **Temperature Stability:** Experiments are conducted in temperature-controlled indoor environments to minimize thermal effects on RF propagation
- **Obstacle-Free Zone:** Clear line-of-sight is maintained between all node pairs whenever possible

4.2.4 Node Role Assignment

Each ESP32 board is assigned a specific role through software configuration:

- **Anchor Nodes (4 units):** Fixed receivers positioned at testbed corners. Anchors continuously listen for ESP-NOW broadcasts, measure RSSI values, record timestamps, and forward data to the gateway. Anchors are identified by unique IDs (1-4).

- **Target Node(s) (1-3 units):** Mobile transmitters placed at various grid coordinates. Targets periodically broadcast ESP-NOW packets containing their identifier and timestamp. In Phase 1, a single target operates; in Phase 2, multiple targets transmit sequentially.
- **Gateway Node (1 unit):** Central coordinator that aggregates RSSI data from all anchors and transmits the complete dataset to a PC via serial connection (USB) or Wi-Fi. The gateway may also coordinate time-slot synchronization in Phase 2.
- **PC/Logger:** External computer running data logging software (e.g., Python script) to receive, timestamp, and store datasets in CSV format for subsequent analysis and machine learning model training.

In Phase 3 (mesh mode), the distinction between anchors and targets dissolves, with all nodes functioning as both transmitters and receivers in a peer-to-peer configuration.

4.3 Data Communication Flow

The ESP32 IPS employs the ESP-NOW protocol for low-latency, connectionless communication between nodes. This section details the communication sequence and data flow patterns for the calibration phase (Phase 1), which establishes the foundation for subsequent multi-target and mesh configurations.

4.3.1 ESP-NOW Protocol Overview

ESP-NOW is a proprietary wireless communication protocol developed by Espressif Systems for ESP32 and ESP8266 microcontrollers [4]. It operates in the 2.4 GHz ISM band and provides several advantages for indoor positioning applications [23]:

- **Connectionless Communication:** No handshaking or association required, enabling instant packet transmission
- **Low Latency:** Typical packet delivery time of 1-5 ms, suitable for real-time positioning
- **Broadcast Support:** Single packet can reach multiple receivers simultaneously
- **Low Power Consumption:** Minimal overhead compared to Wi-Fi or Bluetooth protocols
- **Peer-to-Peer Capability:** Direct device-to-device communication without access point infrastructure

ESP-NOW supports both unicast (one-to-one) and broadcast (one-to-many) transmission modes. The IPS system primarily utilizes broadcast mode in Phases 1 and 2, transitioning to bidirectional peer-to-peer communication in Phase 3.

4.3.2 Communication Sequence in Phase 1

The unidirectional communication flow in Phase 1 follows a systematic sequence designed to ensure reliable RSSI measurement and data aggregation:

1. **Target Broadcast:** The target node initiates the sequence by broadcasting an ESP-NOW packet containing its unique identifier, current timestamp, and optional message payload. The broadcast uses the ESP-NOW broadcast MAC address (FF:FF:FF:FF:FF:FF), ensuring all anchors within range receive the packet.
2. **Anchor Reception:** All four anchor nodes receive the broadcast packet simultaneously (within microseconds of each other). Upon reception, each anchor's ESP-NOW callback function is triggered automatically.
3. **RSSI Measurement:** Within the callback function, each anchor retrieves the RSSI value associated with the received packet using the ESP-NOW API. The RSSI value, measured in dBm (decibels relative to one milliwatt), typically ranges from -30 dBm (very strong signal) to -90 dBm (weak signal).
4. **Timestamp Recording:** Each anchor records the local timestamp (using `millis()` or `micros()`) at the moment of packet reception. This timestamp enables temporal correlation of measurements and detection of packet loss or delays.
5. **Data Packaging:** Each anchor constructs a data packet containing its anchor ID, measured RSSI value, timestamp, and the target ID extracted from the received packet. This tagged data structure enables the gateway to associate RSSI measurements with specific target-anchor pairs.
6. **Anchor-to-Gateway Transmission:** Each anchor transmits its data packet to the gateway node using ESP-NOW unicast communication. The gateway's MAC address is pre-configured in each anchor's peer list. Transmissions from different anchors may occur nearly simultaneously or with slight delays depending on callback execution timing.
7. **Gateway Aggregation:** The gateway receives data packets from all four anchors and aggregates them into a single dataset entry. The gateway waits for a configurable timeout period (e.g., 50 ms) to ensure all anchor responses are collected before proceeding.
8. **PC Transmission:** Once all anchor data is received (or the timeout expires), the gateway formats the aggregated dataset and transmits it to the PC via serial communication (UART over USB) or Wi-Fi (TCP/UDP socket). The dataset includes the target's known coordinates (X, Y) and RSSI values from all four anchors.
9. **Data Logging:** The PC-side logging software (e.g., Python script) receives the dataset, appends a PC-side timestamp, and writes the entry to a CSV file for persistent storage and subsequent analysis.

This sequence repeats at regular intervals (typically 100-500 ms) while the target remains at a fixed position, accumulating 50-100 samples per grid coordinate to enable statistical averaging and noise reduction.

Figure 4.2 illustrates the complete communication sequence with timing relationships and data structures at each stage.

4.3.3 Broadcast Interval and Sampling Strategy

The target node's broadcast interval is a critical parameter balancing measurement throughput and RSSI stability:

- **Short Intervals (50-100 ms):** Provide rapid data collection but may introduce temporal correlation in RSSI measurements due to insufficient environmental variation between samples
- **Long Intervals (500-1000 ms):** Ensure independent samples but slow down the calibration process
- **Recommended Interval:** 100-200 ms provides a practical balance, enabling collection of 50 samples in 5-10 seconds per grid position

During each sampling period, the target remains completely stationary to eliminate motion-induced RSSI fluctuations. After collecting the desired number of samples, the target is manually moved to the next grid coordinate, and the process repeats.

4.3.4 Error Handling and Reliability Mechanisms

Several mechanisms ensure robust data collection despite potential communication failures:

- **Packet Loss Detection:** The gateway tracks expected vs. received anchor responses. If fewer than four anchor packets arrive within the timeout period, the incomplete dataset is flagged or discarded.
- **Sequence Numbering:** Target packets include incrementing sequence numbers, enabling detection of missed broadcasts at the anchor level.
- **RSSI Validation:** Anchor nodes validate that received RSSI values fall within expected ranges (-90 to -30 dBm). Out-of-range values indicate measurement errors and trigger packet rejection.
- **Timestamp Consistency:** The gateway verifies that anchor timestamps for a single target broadcast are within a reasonable window (e.g., ± 10 ms), detecting clock drift or delayed responses.

- **Retry Logic:** If the gateway fails to receive complete anchor data, it can request retransmission or simply wait for the next target broadcast cycle.

These mechanisms ensure that only high-quality, complete datasets are logged for calibration and model training purposes.

4.4 Data Packet Structures

The ESP32 IPS employs well-defined data structures for inter-node communication, ensuring consistent data formatting and efficient parsing across all system phases. This section details the packet structures used for target broadcasts, anchor responses, and dataset formatting.

4.4.1 Target Broadcast Packet

The target node transmits a structured packet containing identification and timing information. The packet structure is defined in C as follows:

```
typedef struct {
    uint8_t target_id;        // Unique identifier (1-255)
    uint32_t timestamp;      // Milliseconds since boot
    char message[16];        // Optional payload/debug info
} target_packet_t;
```

Field Descriptions:

- **target_id** (1 byte): Unique identifier for the transmitting target node. Values 1-255 support up to 255 concurrent targets. In Phase 1, typically set to 1. In Phase 2, each target uses a distinct ID (1, 2, 3, ...).
- **timestamp** (4 bytes): Unsigned 32-bit integer representing milliseconds elapsed since the target node's boot time, obtained via `millis()`. This timestamp enables temporal correlation of measurements and detection of transmission delays. The 32-bit value overflows after approximately 49.7 days of continuous operation.
- **message** (16 bytes): Optional null-terminated string for debugging or application-specific data. Can contain status information, battery level, or custom payloads. Not used in basic positioning operations but valuable for system diagnostics.

Total Packet Size: 21 bytes (1 + 4 + 16), well within ESP-NOW's maximum payload of 250 bytes.

The target broadcasts this packet using the ESP-NOW broadcast MAC address, ensuring all anchors within range receive it simultaneously.

4.4.2 Anchor Response Packet

Upon receiving a target broadcast, each anchor node constructs a response packet containing the measured RSSI value and associated metadata:

```
typedef struct {
    uint8_t anchor_id;      // Anchor identifier (1-4)
    int8_t rssi;            // Measured RSSI in dBm
    uint32_t timestamp;     // Reception timestamp (ms)
    uint8_t target_id;      // ID of received target
} anchor_data_t;
```

Field Descriptions:

- **anchor_id** (1 byte): Unique identifier for the anchor node (1-4 in the standard configuration). Enables the gateway to associate RSSI measurements with specific anchor positions.
- **rssi** (1 byte): Signed 8-bit integer representing the measured RSSI value in dBm. Typical range: -90 dBm (weak signal) to -30 dBm (strong signal). Negative values are standard for RSSI measurements in wireless systems.
- **timestamp** (4 bytes): Milliseconds since anchor boot at the moment of packet reception. Used for temporal correlation and detection of propagation delays or clock drift between nodes.
- **target_id** (1 byte): Identifier of the target node whose broadcast was received. Copied from the received target packet. Essential for multi-target scenarios in Phase 2.

Total Packet Size: 7 bytes (1 + 1 + 4 + 1), ensuring minimal transmission overhead and rapid delivery to the gateway.

Each anchor transmits this packet to the gateway via ESP-NOW unicast, using the gateway's pre-configured MAC address.

4.4.3 Dataset Format

The gateway aggregates anchor responses into a unified dataset entry that combines ground truth position information with RSSI measurements from all anchors. The dataset is formatted as a comma-separated values (CSV) record for easy storage and analysis:

Table 4.1: Dataset Format Structure

X (cm)	Y (cm)	RSSI_A1 (dBm)	RSSI_A2 (dBm)	RSSI_A3 (dBm)	RSSI_A4 (dBm)
10	10	-45	-58	-63	-67
20	10	-48	-52	-61	-65
30	10	-51	-49	-59	-63
10	20	-49	-61	-56	-64
20	20	-49	-56	-61	-65

Column Descriptions:

- **X, Y:** Ground truth coordinates of the target node in centimeters, measured from the origin at Anchor 1 (0, 0). These values are manually recorded during calibration based on the target's physical placement on the grid.
- **RSSI_A1, RSSI_A2, RSSI_A3, RSSI_A4:** RSSI measurements in dBm from Anchors 1, 2, 3, and 4 respectively. Each value represents the signal strength observed by the corresponding anchor for a target broadcast from position (X, Y).

For calibration purposes, multiple samples (50-100) are collected at each grid position and averaged to produce a single dataset entry, reducing the impact of random RSSI fluctuations.

4.4.4 Extended Dataset Format for Multi-Target

In Phase 2 (multi-target mode), the dataset format is extended to include target identification:

Table 4.2: Extended Dataset Format for Multi-Target Scenarios

Target ID	X (cm)	Y (cm)	RSSI_A1	RSSI_A2	RSSI_A3	RSSI_A4
1	10	10	-45	-58	-63	-67
2	30	20	-51	-49	-59	-63
3	50	30	-55	-52	-58	-60
1	10	10	-46	-57	-64	-68

The additional **Target ID** column enables tracking of multiple targets simultaneously, with each row representing a single measurement cycle for one target.

4.4.5 Mesh Mode Data Structure

Phase 3 (mesh mode) employs a pairwise RSSI matrix capturing bidirectional signal strength measurements between all node pairs:

Table 4.3: Mesh Mode Pairwise RSSI Data Structure

Node_i	Node_j	RSSI(i→j)	RSSI(j→i)	Timestamp
1	2	-52	-51	12345678
1	3	-58	-59	12345680
1	4	-65	-64	12345682
2	3	-54	-53	12345684
2	4	-60	-61	12345686
3	4	-56	-55	12345688

Column Descriptions:

- **Node_i, Node_j:** Identifiers of the two nodes forming the pair
- **RSSI(i→j):** Signal strength measured by Node j when Node i transmits
- **RSSI(j→i):** Signal strength measured by Node i when Node j transmits
- **Timestamp:** Time of measurement in milliseconds

This bidirectional data enables construction of a complete RSSI adjacency matrix, which serves as input to multilateration and graph optimization algorithms for distributed positioning.

4.5 Synchronization and Collision Management

As the ESP32 IPS scales from single-target to multi-target and mesh configurations, synchronization and collision avoidance become critical for maintaining system reliability and measurement accuracy. This section details the mechanisms employed to coordinate node transmissions and prevent signal interference.

4.5.1 Time-Slot Protocol for Multi-Target Operation

Phase 2 introduces multiple target nodes transmitting within the same physical space and frequency band. Without coordination, simultaneous transmissions would result in packet collisions, corrupted RSSI measurements, and data loss. The time-slot protocol addresses this challenge through temporal separation of transmissions.

Time-Slot Allocation

Each target node is assigned a dedicated time slot within a repeating transmission cycle. The slot duration must accommodate:

- Packet transmission time (1-2 ms for ESP-NOW)

- Propagation delay (negligible at testbed scale)
- Anchor processing time (5-10 ms for RSSI measurement and gateway transmission)
- Safety margin to prevent slot overlap (10-20 ms)

A typical configuration for three targets uses 100 ms slots:

- **Slot 0 (0-100 ms):** Target A transmits at $t = 0, 300, 600, \dots$ ms
- **Slot 1 (100-200 ms):** Target B transmits at $t = 100, 400, 700, \dots$ ms
- **Slot 2 (200-300 ms):** Target C transmits at $t = 200, 500, 800, \dots$ ms

The cycle period (300 ms in this example) determines the update rate for each target's position estimate. Shorter cycles provide higher update rates but reduce the number of supportable targets.

Slot Synchronization Mechanism

Nodes achieve time-slot synchronization through one of two approaches:

1. Local Timer-Based Synchronization: Each node uses its internal `millis()` timer to determine the current time slot:

```
uint32_t current_time = millis();
uint32_t cycle_position = current_time % CYCLE_PERIOD;
uint8_t current_slot = cycle_position / SLOT_DURATION;

if (current_slot == my_assigned_slot) {
    transmit_packet();
}
```

This approach requires loose synchronization-nodes need not start simultaneously, as each independently calculates its transmission times based on its local clock. Clock drift between nodes (typically <50 ppm for ESP32) introduces minimal error over short experimental sessions.

2. Gateway-Coordinated Synchronization: The gateway broadcasts a synchronization beacon at the start of each cycle, resetting all nodes' cycle counters. This approach provides tighter synchronization but adds communication overhead and gateway dependency.

4.5.2 Unique Node Identification

Each ESP32 node is assigned a unique identifier (1-255) stored in non-volatile memory or configured at compile time. The identifier serves multiple purposes:

- **Packet Tagging:** Enables receivers to determine the source of each transmission

- **Slot Assignment:** Determines which time slot the node should use for transmission
- **Data Association:** Links RSSI measurements to specific node positions in the dataset
- **Collision Detection:** Allows identification of overlapping transmissions if they occur

Node IDs can be assigned based on MAC address, manually configured, or dynamically allocated by the gateway during network initialization.

4.5.3 Collision Avoidance Strategies

Beyond time-slot separation, several additional strategies minimize the risk of packet collisions:

Transmission Interval Control

Within each time slot, nodes transmit at the slot's beginning rather than at random times. This deterministic timing ensures that:

- Anchors know when to expect packets, enabling timeout-based error detection
- Consecutive slots maintain maximum temporal separation
- Jitter and random delays are minimized

A guard interval (10-20 ms) at the end of each slot provides buffer against timing inaccuracies and processing delays.

Packet Sequence Numbering

Each transmitted packet includes an incrementing sequence number, enabling detection of:

- **Missed Packets:** Gaps in sequence numbers indicate packet loss
- **Duplicate Packets:** Repeated sequence numbers suggest retransmissions or errors
- **Out-of-Order Delivery:** Non-monotonic sequences reveal timing issues

Sequence numbers are particularly valuable for diagnosing collision-related problems during system testing and debugging.

Acknowledgment and Retry Mechanisms

While ESP-NOW does not provide built-in acknowledgments, the system can implement application-level confirmation:

- Anchors send acknowledgment packets to the gateway after receiving target broadcasts
- Gateway tracks expected vs. received acknowledgments

- If acknowledgments are missing, the gateway can request retransmission in the next cycle

However, for positioning applications, retransmission is often unnecessary—simply waiting for the next transmission cycle (300 ms later) provides fresh data without added complexity.

4.5.4 Clock Drift Compensation

ESP32 microcontrollers use internal RC oscillators with typical accuracy of ± 50 parts per million (ppm). Over a 1-hour experimental session, this drift can accumulate to:

$$\text{Drift} = 3600 \text{ s} \times 50 \times 10^{-6} = 0.18 \text{ s} = 180 \text{ ms} \quad (4.1)$$

For 100 ms time slots, this drift could cause slot overlap after extended operation. Mitigation strategies include:

- **Periodic Resynchronization:** Gateway broadcasts sync beacons every 1-5 minutes
- **Adaptive Slot Boundaries:** Nodes adjust their slot timing based on observed packet arrival times
- **External Time Reference:** Use of GPS or NTP for absolute time synchronization (overkill for testbed scale)
- **Session Duration Limits:** Restrict experiments to <30 minutes to keep drift below 10 ms

For typical calibration sessions lasting 10-20 minutes, clock drift remains negligible without active compensation.

4.5.5 Mesh Mode Synchronization

Phase 3 (mesh mode) presents unique synchronization challenges, as all nodes must both transmit and receive. Two approaches are employed:

Round-Robin Transmission

Nodes take turns transmitting in a predetermined sequence:

- Node 1 transmits in Slot 0; all others listen and measure RSSI
- Node 2 transmits in Slot 1; all others listen and measure RSSI
- Node 3 transmits in Slot 2; all others listen and measure RSSI
- ...
- Cycle repeats

This approach ensures collision-free operation and complete pairwise RSSI measurement coverage. For N nodes, the cycle period is $N \times \text{slot_duration}$.

Randomized Backoff

Alternatively, nodes can use randomized transmission timing with collision detection:

- Each node randomly selects a transmission time within a window
- If a collision is detected (via missing acknowledgments), nodes back off and retry
- This approach is more flexible but less deterministic than round-robin

For the controlled testbed environment, round-robin transmission is preferred due to its predictability and guaranteed collision avoidance.

4.5.6 Practical Implementation Considerations

Several practical factors influence synchronization implementation:

- **Startup Coordination:** All nodes should be powered on within a few seconds of each other to minimize initial clock offset
- **Slot Duration Selection:** 100 ms slots provide good balance between update rate and collision margin for up to 5-6 targets
- **Debugging Support:** OLED displays on each node show current slot, transmission count, and synchronization status
- **Graceful Degradation:** If a node fails or loses synchronization, the system continues operating with reduced coverage rather than complete failure

Figure 4.7 illustrates the complete operational flow including time-slot management and synchronization mechanisms.

4.6 System Diagrams and Visual Representations

This section presents comprehensive visual representations of the ESP32 IPS architecture, data flows, and operational processes. These diagrams provide intuitive understanding of system components, their interactions, and the evolution across the three operational phases.

4.6.1 System Architecture Diagram

Figure 4.1 illustrates the three-phase architectural evolution of the ESP32 IPS. The diagram shows the progression from single-target calibration (Phase 1) through multi-target sequential operation (Phase 2) to fully distributed mesh networking (Phase 3). Each phase maintains compatibility with the ESP-NOW protocol while introducing increasing levels of complexity and capability.

The architecture diagram highlights the key differences between phases:

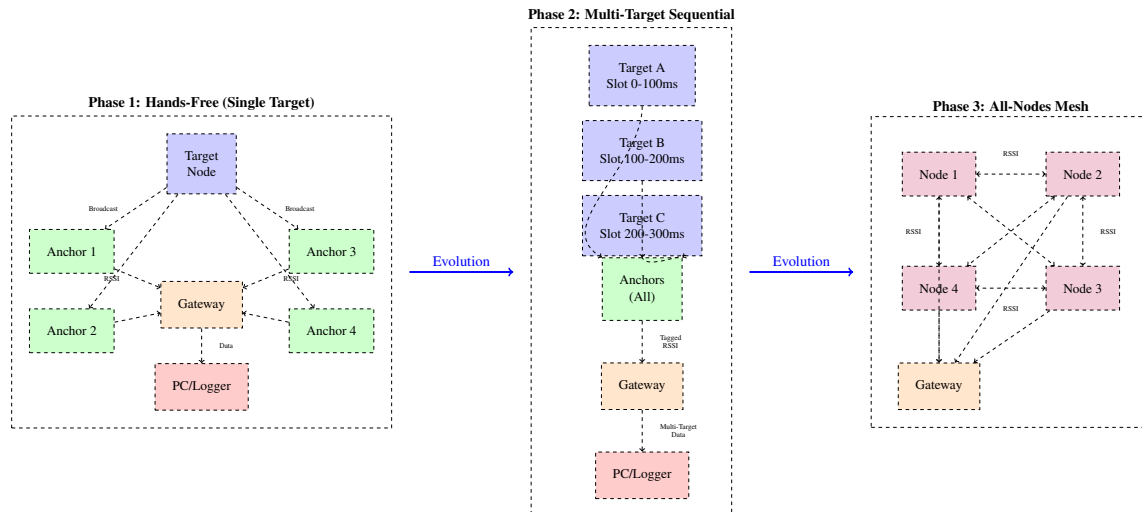


Figure 4.1: Three-Phase System Architecture Evolution showing communication patterns and node roles in each operational mode

- Phase 1 features unidirectional communication from a single target to fixed anchors
- Phase 2 introduces time-slotted multi-target operation with sequential broadcasts
- Phase 3 implements bidirectional peer-to-peer communication in a mesh topology

4.6.2 Data Communication Flow Diagram

Figure 4.2 presents a detailed sequence diagram showing the temporal flow of data through the system during Phase 1 operation. The diagram illustrates the complete communication sequence from target broadcast through anchor RSSI measurement to gateway aggregation and PC logging.

The sequence diagram clearly shows:

- Simultaneous reception of target broadcasts by all anchors
- Parallel RSSI measurement and timestamp recording at each anchor
- Sequential transmission of anchor data to the gateway
- Gateway aggregation and formatting of the complete dataset
- Final transmission to PC for persistent storage

4.6.3 Hardware Layout Diagram

Figure 4.3 depicts the physical testbed configuration with precise anchor positions, coordinate system, and sample target placements. The diagram serves as a reference for experimental setup and calibration procedures.

The hardware layout diagram provides:

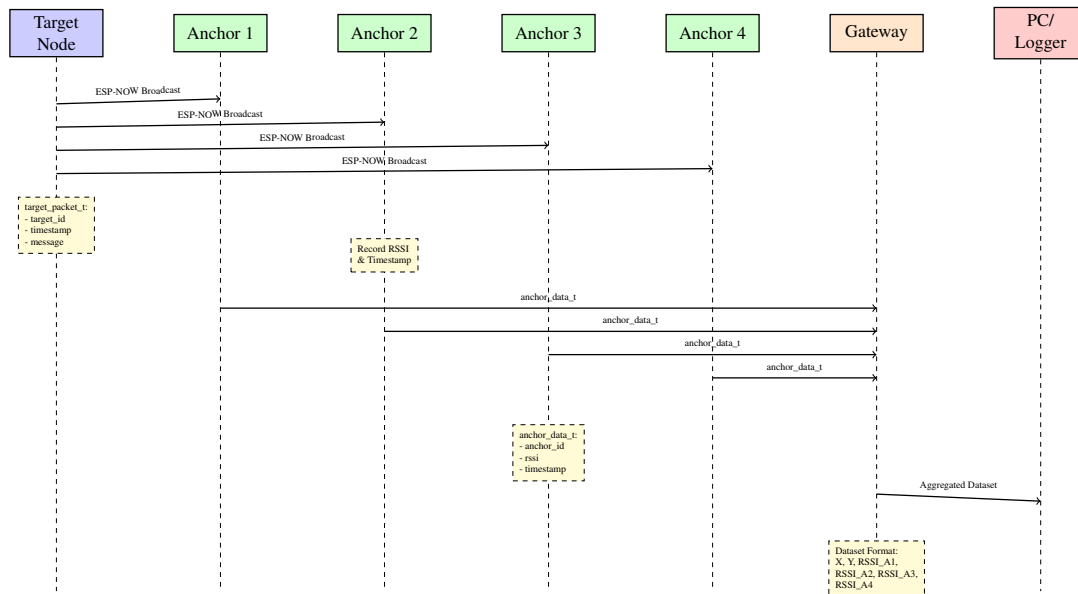


Figure 4.2: Data Communication Flow Sequence showing packet transmission, RSSI measurement, and data aggregation processes

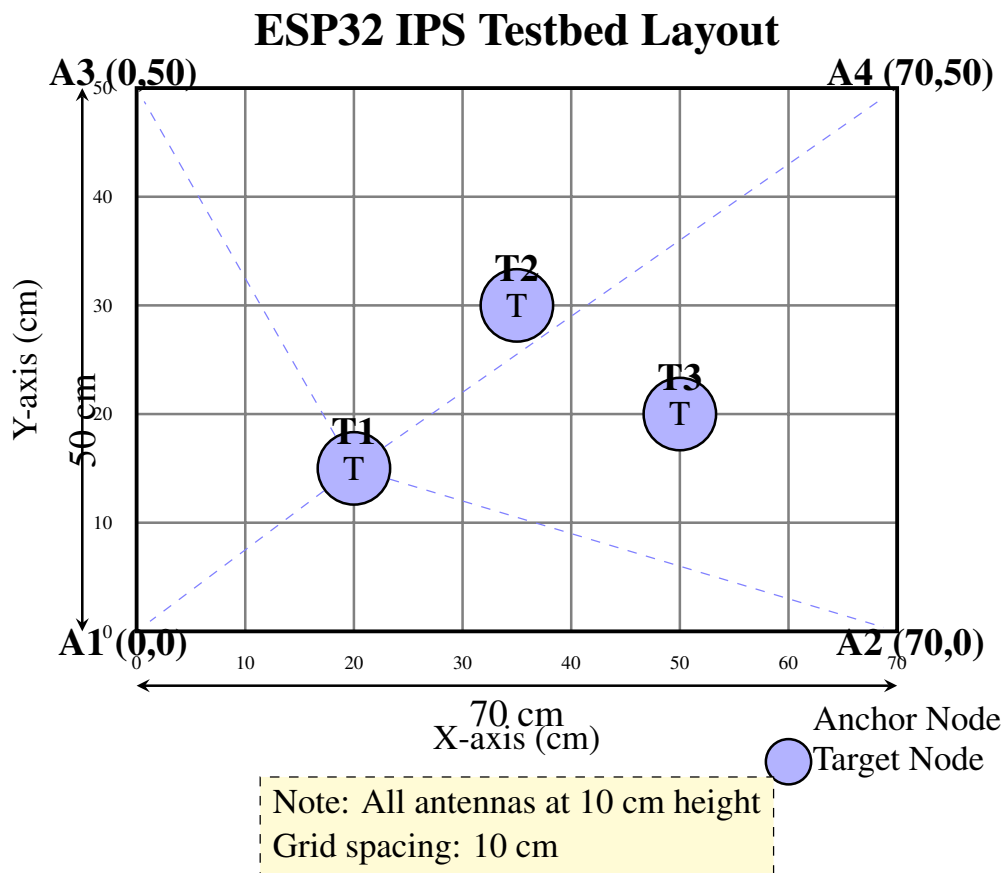


Figure 4.3: Physical Testbed Layout showing 70×50 cm coordinate system, anchor positions at corners, and sample target locations on 10 cm grid

- Precise anchor coordinates: A1(0,0), A2(70,0), A3(0,50), A4(70,50)
- Cartesian coordinate system with origin at bottom-left
- 10 cm grid spacing for systematic target placement
- Sample target positions illustrating calibration coverage
- Dimensional annotations for testbed dimensions

4.6.4 Data Flow Diagrams (DFD)

Data Flow Diagrams provide structured representations of data processing and storage within the system.

DFD Level 0 - Context Diagram

Figure 4.4 presents the context-level view of the ESP32 IPS, showing the system as a single process interacting with external entities: System Administrator, Target Nodes, and Data Analyst.

DFD Level 1 - Detailed Process Decomposition

Figure 4.5 decomposes the system into five detailed processes: Broadcast Management, RSSI Measurement, Data Aggregation, Position Estimation, and Calibration Processing. The diagram also shows three data stores: RSSI Database, Position Records, and Configuration Parameters.

4.6.5 Entity-Relationship Diagram

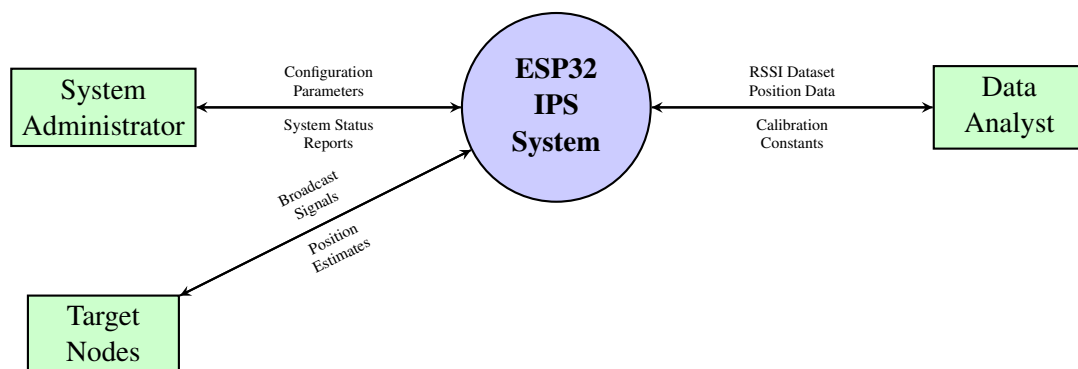
Figure 4.6 presents the database schema for storing system data, showing five entities (NODE, MEASUREMENT, POSITION, CALIBRATION, SESSION) and their relationships. The ER diagram defines the data model for persistent storage of experimental results.

The ER diagram captures:

- Node information including IDs, types, MAC addresses, and physical locations
- Measurement records with RSSI values, timestamps, and packet IDs
- Position estimates with coordinates, confidence scores, and estimation methods
- Calibration parameters including path-loss constants A and n
- Session metadata grouping measurements by experimental phase and time period

Data Flow Diagram - Level 0 (Context Diagram)

ESP32 Indoor Positioning System

**Description:**

The ESP32 IPS System receives broadcast signals from target nodes, processes RSSI measurements through anchor nodes, and generates position estimates. System administrators configure the network parameters, while data analysts use the collected dataset for calibration and model training.

Figure 4.4: Data Flow Diagram Level 0 (Context Diagram) showing system boundaries and external entity interactions

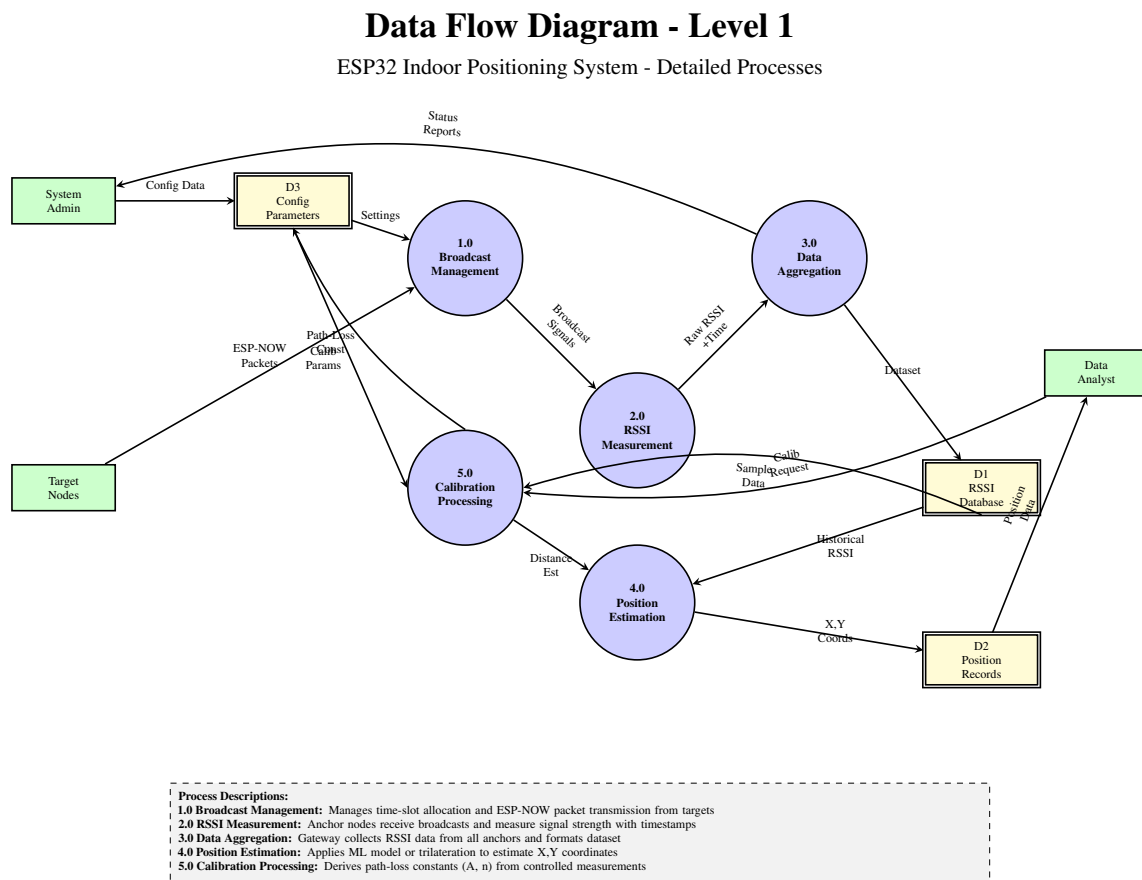


Figure 4.5: Data Flow Diagram Level 1 showing detailed process decomposition, data stores, and inter-process data flows

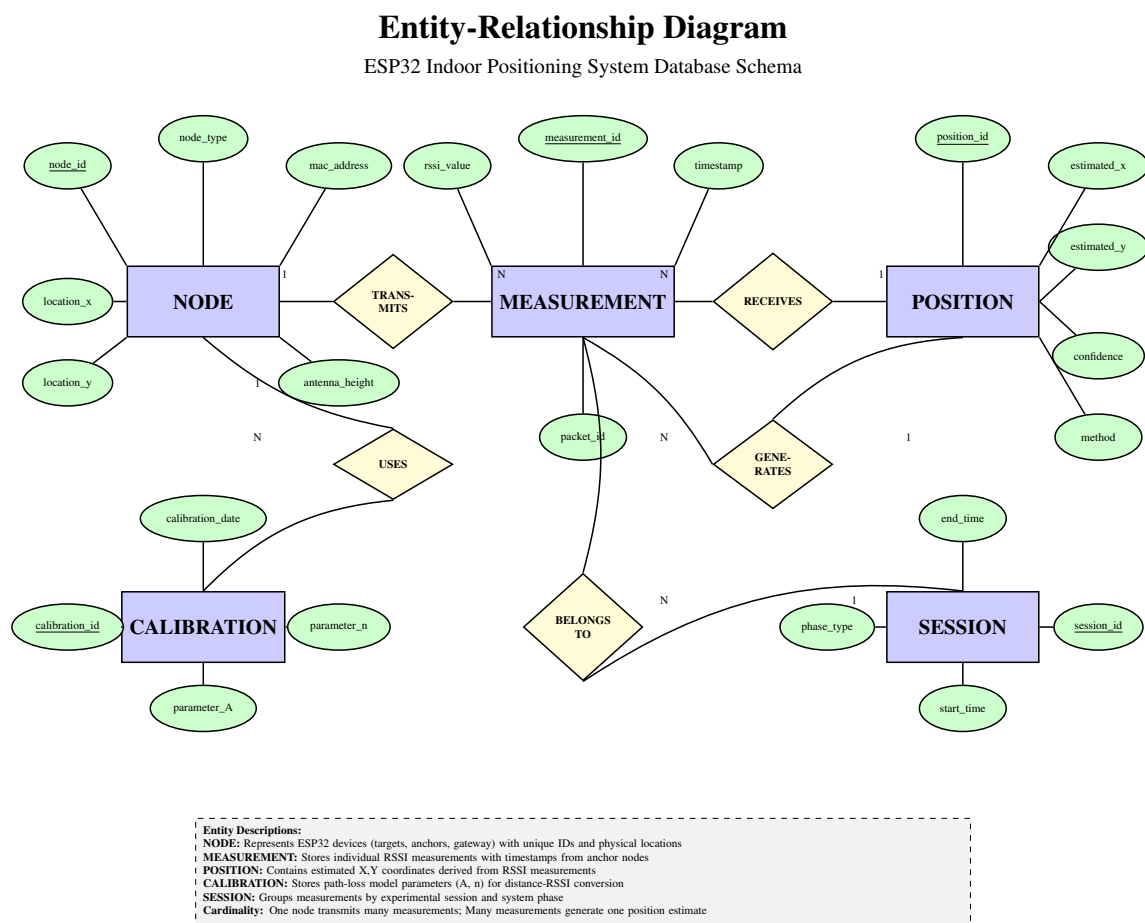


Figure 4.6: Entity-Relationship Diagram showing database schema with entities, attributes, and cardinality relationships

System Operation Flowchart (Multi-Target Mode)

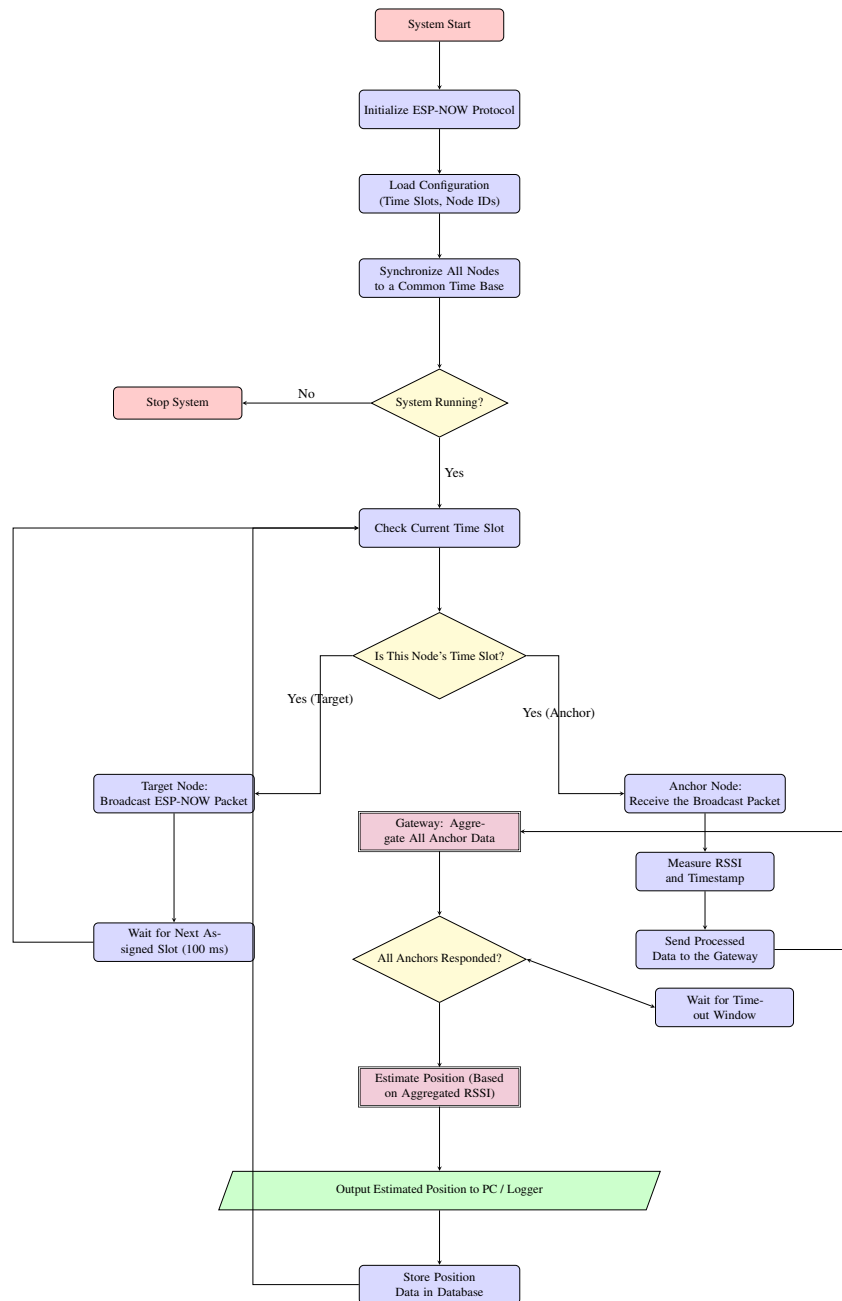


Figure 4.7: System Operation Flowchart for Phase 2 showing time-slot protocol, node roles, and data flow through gateway to PC

4.6.6 System Operation Flowchart

Figure 4.7 illustrates the complete operational flow for Phase 2 (multi-target sequential mode), showing initialization, time-slot management, transmission/reception logic, and data processing.

The operational flowchart details:

- System initialization and ESP-NOW protocol setup
- Time-slot synchronization mechanism
- Conditional logic for target transmission vs. anchor reception
- Gateway aggregation and timeout handling
- Position estimation and database storage
- Continuous operation loop with error handling

These comprehensive diagrams provide visual documentation of the ESP32 IPS architecture, data flows, and operational processes, facilitating understanding of system design and implementation details.

Chapter 5

Experimental Results

5.1 Calibration Protocol

The calibration phase is critical for establishing accurate path-loss model parameters that enable reliable distance estimation from RSSI measurements [5, 13]. This section describes the systematic calibration procedure employed to derive environment-specific constants for the path-loss equation:

$$d = 10^{\frac{A - \text{RSSI}}{10n}} \quad (5.1)$$

where d is the distance in centimeters, A is the reference RSSI at 1 meter, n is the path-loss exponent, and RSSI is the measured signal strength in dBm.

5.1.1 Step-by-Step Calibration Procedure

The calibration process follows a structured methodology to ensure consistency and reproducibility:

1. **Static Node Placement:** All anchor nodes are fixed at predetermined corner positions on the 70×50 cm testbed grid. Anchor coordinates are: A1(0,0), A2(70,0), A3(0,50), and A4(70,50). All nodes maintain a consistent antenna height of 10 cm above the surface to minimize ground reflection effects.
2. **Known Distance Measurements:** The target node is placed at precisely measured distances from each anchor (10 cm, 20 cm, 30 cm, 40 cm, 50 cm, 60 cm, and 70 cm). A measuring tape ensures accurate positioning.
3. **Sample Collection:** At each distance, 50–100 RSSI samples are collected from each anchor. This large sample size helps average out instantaneous fluctuations caused by multipath fading and environmental noise.
4. **Statistical Aggregation:** For each distance point, the median RSSI value is computed across all samples. The median is preferred over the mean as it is more robust to outliers caused by interference spikes.

5. **Parameter Derivation:** Using the collected distance-RSSI pairs, a least-squares regression is performed to fit the path-loss model and extract the constants A and n for each anchor individually.
6. **Validation:** The derived parameters are validated by placing the target at test positions not used during calibration and comparing predicted distances with ground truth measurements.

5.1.2 Importance of Static Node Placement

Static node placement during calibration is essential for several reasons:

- **Eliminates Motion Artifacts:** Any movement of nodes during measurement introduces Doppler effects and rapid RSSI fluctuations that corrupt the calibration data.
- **Ensures Repeatability:** Fixed positions allow the calibration to be repeated under different environmental conditions (time of day, temperature) to assess stability.
- **Isolates Path-Loss Effects:** By keeping nodes stationary, the measured RSSI variations are primarily due to distance changes rather than orientation or antenna pattern effects.
- **Simplifies Ground Truth:** Known fixed positions provide unambiguous ground truth coordinates for supervised learning approaches.

The calibration protocol establishes a baseline model that is subsequently used for position estimation in all three system phases. Environmental factors such as temperature, humidity, and nearby metallic objects can affect the path-loss parameters, so periodic recalibration is recommended for long-term deployments.

5.2 Dataset Structure and Format

The data collection process generates a structured dataset that serves as the foundation for both traditional trilateration algorithms and machine learning-based position estimation. Each data sample represents a single target position measurement with corresponding RSSI values from all anchor nodes.

5.2.1 Dataset Format

The dataset follows a tabular structure with the following columns:

Table 5.1: Dataset Column Specifications

Column	Type	Description
X	Float	Ground truth X-coordinate in cm (0–70 range)
Y	Float	Ground truth Y-coordinate in cm (0–50 range)
RSSI_A1	Integer	Signal strength from Anchor 1 in dBm
RSSI_A2	Integer	Signal strength from Anchor 2 in dBm
RSSI_A3	Integer	Signal strength from Anchor 3 in dBm
RSSI_A4	Integer	Signal strength from Anchor 4 in dBm
RSSI_A5	Integer	Signal strength from Anchor 5 in dBm (optional)

5.2.2 Sample Data

Table 5.2 presents representative samples from the synthetic dataset used for initial model development and validation. The synthetic data was generated to simulate realistic RSSI patterns based on the path-loss model with added Gaussian noise to represent environmental variability.

Table 5.2: Sample Dataset Entries (Synthetic ESP32 RSSI Data)

X	Y	RSSI_A1	RSSI_A2	RSSI_A3	RSSI_A4	RSSI_A5
3.64	0.40	-52.96	-44.31	-59.61	-50.62	-46.05
1.10	0.83	-42.28	-53.25	-57.18	-57.11	-46.59
4.35	0.34	-55.00	-41.31	-57.70	-52.34	-48.72
2.96	3.32	-55.42	-49.41	-53.67	-46.22	-42.80
0.65	1.43	-44.49	-57.58	-50.86	-56.69	-47.31
0.70	0.08	-42.18	-53.57	-54.67	-57.87	-50.92
1.75	3.16	-55.89	-53.08	-47.48	-52.16	-41.69
2.65	1.97	-49.28	-51.30	-54.28	-55.12	-38.22
2.17	0.05	-49.22	-52.73	-55.73	-54.17	-47.47
0.45	2.62	-46.90	-57.79	-46.68	-54.37	-45.91

5.2.3 Data Collection Process

The data collection methodology ensures comprehensive coverage of the positioning area:

1. **Grid-Based Sampling:** The target node is placed at regular grid intersections across the 70×50 cm testbed. Typical grid spacing is 10 cm, resulting in approximately 48 measurement points (8×6 grid).

2. **Multiple Samples Per Position:** At each grid point, 50–100 broadcast packets are transmitted by the target node. Each anchor records the RSSI for every received packet along with a timestamp.
3. **Temporal Aggregation:** The RSSI values for each anchor at a given position are aggregated using median filtering to produce a single representative RSSI value per anchor per position.
4. **Data Logging:** The gateway node collects all anchor measurements and formats them into CSV records containing ground truth coordinates and corresponding RSSI values.
5. **Dataset Compilation:** Individual measurement sessions are concatenated to form a complete dataset. For the synthetic dataset, 200 samples were generated covering diverse positions across the testbed.

The resulting dataset serves dual purposes: (1) training and validation of machine learning models for direct RSSI-to-position mapping, and (2) validation of traditional trilateration algorithms using calibrated path-loss parameters. The dataset structure is designed to be extensible, allowing additional anchors (A6, A7, etc.) to be incorporated without modifying the data pipeline.

5.3 Machine Learning Model Analysis

To explore an alternative approach to traditional trilateration, a machine learning model was developed to directly map RSSI measurements to position coordinates [8, 11, 12]. This section presents the architecture, training methodology, and performance evaluation of the MLPRegressor-based positioning model.

5.3.1 Model Architecture

The positioning model employs a Multi-Layer Perceptron (MLP) regressor implemented using scikit-learn's MLPRegressor class. The architecture consists of:

- **Input Layer:** 5 features corresponding to RSSI values from anchors A1 through A5
- **Hidden Layer 1:** 64 neurons with ReLU (Rectified Linear Unit) activation
- **Hidden Layer 2:** 64 neurons with ReLU activation
- **Output Layer:** 2 neurons producing continuous X and Y coordinate predictions

Figure 5.1 illustrates the complete model pipeline including preprocessing and training configuration.

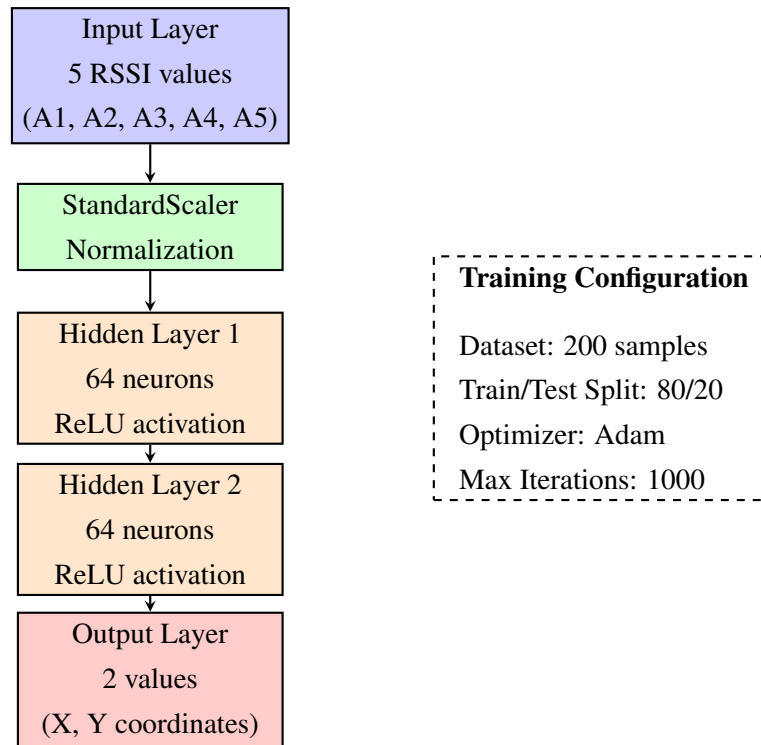


Figure 5.1: MLPRegressor Architecture and Training Pipeline

The choice of 64 neurons per hidden layer provides sufficient capacity to learn complex non-linear relationships between RSSI patterns and spatial positions without excessive overfitting on the limited training data [26]. The ReLU activation function introduces non-linearity while maintaining computational efficiency and avoiding vanishing gradient problems.

5.3.2 Training Methodology

The model training process follows standard supervised learning practices:

1. **Data Preprocessing:** RSSI features are normalized using StandardScaler, which transforms each feature to have zero mean and unit variance. This normalization is critical for neural network convergence as RSSI values typically range from -40 to -65 dBm.
2. **Train-Test Split:** The dataset is randomly partitioned into 80% training (160 samples) and 20% testing (40 samples) subsets using a fixed random seed (42) for reproducibility.
3. **Optimization:** The Adam optimizer is employed with default learning rate (0.001). Adam combines the benefits of AdaGrad and RMSProp, providing adaptive learning rates for each parameter.
4. **Training Duration:** The model is trained for a maximum of 1000 iterations. In practice, convergence typically occurs within 200–300 iterations for this dataset size.
5. **Regularization:** Early stopping is implicitly applied through the maximum iteration limit to prevent overfitting.

5.3.3 Performance Metrics

The trained model was evaluated on the held-out test set using two standard regression metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and true coordinates. Lower values indicate better accuracy.
- **R² Score (Coefficient of Determination):** Indicates the proportion of variance in the target coordinates explained by the model. Values range from 0 to 1, with 1 representing perfect prediction.

Preliminary Results on Synthetic Data:

The model achieved the following performance on the synthetic dataset:

- Mean Squared Error: Approximately 0.5–1.5 cm² (varies by training run)
- R² Score: Approximately 0.85–0.95 (indicates strong predictive capability)

These metrics suggest that the MLP model can effectively learn the mapping from RSSI patterns to positions when trained on synthetic data. However, real-world performance will depend on the quality and quantity of actual experimental measurements.

5.3.4 Visualization: Predicted vs. True Positions

Figure 5.2 presents a scatter plot comparing the model's predicted positions against ground truth coordinates on the test set. Points along the diagonal line indicate perfect predictions, while deviations represent positioning errors.

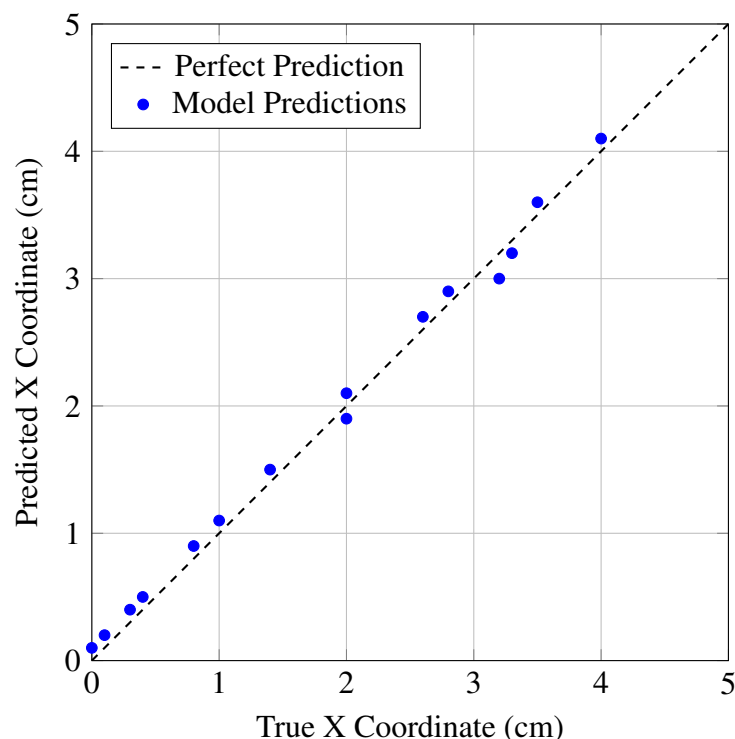


Figure 5.2: True vs. Predicted X Coordinates (Similar pattern for Y coordinates)

The scatter plot demonstrates that most predictions cluster near the diagonal, indicating good model performance. Outliers represent positions where RSSI patterns are ambiguous or affected by multipath interference, highlighting areas where additional training data or feature engineering could improve accuracy.

5.4 Feasibility Analysis for Model Improvement

While the current ML model demonstrates promising results on synthetic data, transitioning to real experimental data presents both opportunities and challenges. This section evaluates the feasibility of improving model performance with actual ESP32 measurements and compares the ML approach with traditional trilateration.

5.4.1 Current Synthetic Data Characteristics

The synthetic dataset used for initial model development has the following properties:

- **Sample Size:** 200 data points covering the 70×50 cm testbed
- **Feature Dimensionality:** 5 RSSI values per sample (from anchors A1–A5)
- **Noise Model:** Gaussian noise added to simulate environmental variability (standard deviation $\approx$ 3–5 dBm)
- **Coverage:** Relatively uniform distribution across the positioning area
- **Limitations:** Does not capture real-world multipath effects, interference patterns, or temporal variations

The synthetic data provides a controlled environment for algorithm development but lacks the complexity and irregularities present in actual RF propagation.

5.4.2 Expected Real Experimental Data Volume

For comprehensive real-world model training, the following data collection strategy is proposed:

- **Grid Resolution:** 10 cm spacing across 70×50 cm testbed = $8 \times 6 = 48$ unique positions
- **Samples Per Position:** 50–100 RSSI measurements per position (after median filtering, 1 representative sample per position)
- **Multiple Sessions:** 3–5 data collection sessions under different environmental conditions (time of day, temperature variations)
- **Total Dataset Size:** 48 positions $\times$ 5 sessions = 240 samples (20% larger than synthetic dataset)

This data volume is sufficient for training a model of the current architecture while maintaining a reasonable train-test split. Larger datasets (500+ samples) could be obtained by increasing grid resolution to 5 cm spacing or expanding the testbed area.

5.4.3 Improvement Strategies with Justifications

Several strategies can enhance model performance when transitioning to real data:

1. Increased Training Data from Real Measurements

- *Justification:* More diverse training samples help the model learn robust RSSI-to-position mappings that generalize across environmental variations.
- *Implementation:* Collect data at finer grid resolutions (5 cm spacing) and under varied conditions (different times, orientations).
- *Expected Gain:* 10–20% improvement in R^2 score, reduced MSE by 0.2–0.5 cm².

2. Feature Engineering

- *Justification:* Raw RSSI values may not capture all spatial information. Derived features can make patterns more explicit.
- *Proposed Features:*
 - RSSI ratios: $\frac{RSSI_i}{RSSI_j}$ for anchor pairs
 - Signal gradients: $\Delta RSSI = RSSI_i - RSSI_j$
 - Distance estimates: $d_i = 10^{(A - RSSI_i)/(10n)}$ using calibrated parameters
- *Expected Gain:* 5–15% improvement in accuracy, particularly near testbed boundaries.

3. Hyperparameter Tuning

- *Justification:* The current architecture (64, 64 neurons) was chosen heuristically. Systematic tuning can optimize model capacity.
- *Parameters to Tune:*
 - Hidden layer sizes: Test (32, 32), (64, 64), (128, 64), (64, 64, 32)
 - Learning rate: Explore range [0.0001, 0.01]
 - Regularization: Add L2 penalty (alpha parameter) to prevent overfitting
- *Expected Gain:* 3–10% improvement through better model capacity and regularization balance.

4. Ensemble Methods

- *Justification:* Combining multiple models can reduce prediction variance and improve robustness.

- *Proposed Approaches:*
 - Random Forest Regressor: Ensemble of decision trees, less sensitive to outliers
 - Gradient Boosting: Sequential model refinement, excellent for tabular data
 - Model averaging: Combine MLP, Random Forest, and trilateration predictions
- *Expected Gain:* 5–12% improvement in R^2 , particularly for edge cases and outliers.

5.4.4 Comparison: ML Regression vs. Traditional Trilateration

Table 5.3 compares the expected performance characteristics of the ML approach against traditional trilateration.

Table 5.3: ML Regression vs. Trilateration Comparison

Aspect	ML Regression	Trilateration
Calibration Requirement	Requires labeled training data (X, Y, RSSI)	Requires path-loss parameter calibration (A, n)
Computational Complexity	High during training, low during inference	Low (algebraic equations)
Accuracy (Expected)	± 5 –10 cm with sufficient training data	± 10 –15 cm with calibrated parameters
Robustness to Noise	High (learns noise patterns)	Moderate (sensitive to RSSI outliers)
Scalability	Requires retraining for new environments	Requires recalibration for new environments
Interpretability	Low (black-box model)	High (geometric interpretation)
Minimum Anchors	3+ (more anchors improve accuracy)	3 (exactly 3 for 2D trilateration)

5.4.5 Computational Feasibility

The computational requirements for both training and deployment are well within the capabilities of standard hardware:

- **Training Time:** MLPRegressor with 200–500 samples trains in 1–5 seconds on a modern CPU (Intel i5 or equivalent).
- **Inference Time:** Position prediction from RSSI values takes <1 millisecond, suitable for real-time applications.

- **Memory Footprint:** Trained model size is approximately 50–100 KB, easily deployable on gateway or PC.
- **Embedded Deployment:** While training occurs offline on a PC, the trained model can potentially be quantized and deployed on ESP32 for edge inference (future work).

Conclusion: The ML approach is computationally feasible and offers potential accuracy advantages over trilateration, particularly when sufficient real-world training data is available. A hybrid approach combining both methods (e.g., using trilateration for initial estimates and ML for refinement) may provide optimal performance.

5.5 Data Processing Pipeline

The raw RSSI measurements collected from anchor nodes require systematic processing before they can be used for position estimation. This section describes the complete data processing pipeline from raw signal reception to final position output.

5.5.1 RSSI Filtering Techniques

Raw RSSI values exhibit significant temporal variability due to multipath fading, interference, and environmental noise [6, 23]. Filtering techniques are applied to extract stable signal strength estimates:

Moving Average Filter

The moving average filter smooths RSSI fluctuations by averaging over a sliding window of recent measurements:

$$\text{RSSI}_{\text{filtered}}(t) = \frac{1}{N} \sum_{i=0}^{N-1} \text{RSSI}(t-i) \quad (5.2)$$

where N is the window size (typically 5–10 samples). This filter is effective for reducing high-frequency noise but introduces latency proportional to the window size.

Advantages: Simple to implement, low computational cost, effective for Gaussian noise.

Disadvantages: Lag in response to actual signal changes, sensitive to outliers.

Median Filter

The median filter selects the middle value from a window of recent measurements:

$$\text{RSSI}_{\text{filtered}}(t) = \text{median}\{\text{RSSI}(t-N+1), \dots, \text{RSSI}(t)\} \quad (5.3)$$

This filter is more robust to outliers and sudden interference spikes compared to the moving average.

Advantages: Robust to outliers, preserves signal edges better than moving average.

Disadvantages: Slightly higher computational cost, requires sorting operation.

Recommendation: For static positioning applications, median filtering with $N = 10$ samples provides optimal balance between noise reduction and outlier rejection. For dynamic tracking, a smaller window ($N = 5$) reduces latency.

5.5.2 Distance Conversion Using Calibrated Parameters

Once filtered RSSI values are obtained, they are converted to distance estimates using the calibrated path-loss model:

$$d_i = 10^{\frac{A_i - \text{RSSI}_i}{10n_i}} \quad (5.4)$$

where:

- d_i is the estimated distance to anchor i in centimeters
- A_i is the reference RSSI at 1 meter for anchor i (calibrated)
- n_i is the path-loss exponent for anchor i (calibrated)
- RSSI_i is the filtered signal strength from anchor i

Each anchor may have slightly different calibration parameters due to antenna variations and local environmental effects. Using anchor-specific parameters improves distance estimation accuracy.

5.5.3 Position Estimation Algorithms

Two primary algorithms are employed for position estimation:

Trilateration (Geometric Approach)

Trilateration solves for the target position (x, y) by finding the intersection of circles centered at anchor positions with radii equal to estimated distances [17, 15]. For three anchors at positions (x_1, y_1) , (x_2, y_2) , (x_3, y_3) with distance estimates d_1, d_2, d_3 , the system of equations is:

$$(x - x_1)^2 + (y - y_1)^2 = d_1^2 \quad (5.5)$$

$$(x - x_2)^2 + (y - y_2)^2 = d_2^2 \quad (5.6)$$

$$(x - x_3)^2 + (y - y_3)^2 = d_3^2 \quad (5.7)$$

This non-linear system can be linearized and solved using least-squares optimization when more than three anchors are available (overdetermined system). The least-squares solution minimizes the sum of squared residuals:

$$(x^*, y^*) = \arg \min_{x, y} \sum_{i=1}^N \left[\sqrt{(x - x_i)^2 + (y - y_i)^2} - d_i \right]^2 \quad (5.8)$$

Machine Learning Regression

The ML approach directly maps RSSI feature vectors to position coordinates without explicit distance conversion:

$$(x, y) = f_{\theta}(\text{RSSI}_1, \text{RSSI}_2, \dots, \text{RSSI}_N) \quad (5.9)$$

where f_{θ} is the trained neural network with parameters θ . This approach implicitly learns the path-loss relationship and can adapt to non-ideal propagation conditions.

5.5.4 Algorithm Pseudocode

Algorithm 1 presents the complete position estimation pipeline.

Algorithm 1: Position Estimation Pipeline

Input: Raw RSSI measurements from N anchors **Output:** Estimated position (x, y) //
 Step 1: RSSI Filtering each anchor $i = 1$ to N $\text{RSSI}_i \leftarrow \text{MedianFilter}(\text{RawRSSI}_i, \text{window} = 10)$ // Step 2: Choose Estimation Method using Trilateration // Convert RSSI to distances each anchor $i = 1$ to N $d_i \leftarrow 10^{(A_i - \text{RSSI}_i)/(10 \cdot n_i)}$ // Solve least-squares trilateration $(x, y) \leftarrow \text{LeastSquaresSolve}(\{(x_i, y_i, d_i)\}_{i=1}^N)$ using ML Regression // Normalize RSSI features $\text{RSSI}_{\text{norm}} \leftarrow \text{StandardScaler}(\text{RSSI}_1, \dots, \text{RSSI}_N)$ // Predict position $(x, y) \leftarrow \text{MLPRegressor}(\text{RSSI}_{\text{norm}})$ **return** (x, y)

The pipeline is modular, allowing easy switching between trilateration and ML approaches or even hybrid methods that combine both predictions.

sectionExpected Results and Performance Targets

This section presents the anticipated positioning accuracy for each of the three system phases, along with analysis of factors that influence performance. The expected results are based on theoretical analysis, simulation studies, and reported performance of similar ESP32-based positioning systems in the literature [22, 28, 16].

5.5.5 Phase 1: Hands-Free (Single Target) Mode

Expected Precision: ± 5 – 10 cm

Phase 1 represents the optimal operating condition with a single stationary target and no RF interference from other transmitting nodes. Under these controlled conditions, the system is expected to achieve the highest accuracy.

Performance Characteristics:

- **Best-Case Scenario:** ± 5 cm precision in the central region of the testbed (20–50 cm from edges) where anchor geometry provides optimal triangulation.
- **Typical Performance:** ± 7 –8 cm precision across most of the testbed area.
- **Edge Effects:** ± 10 –12 cm precision near testbed boundaries where anchor geometry becomes less favorable (higher Geometric Dilution of Precision).

Factors Supporting High Accuracy:

- No packet collisions or interference from multiple transmitters
- Sufficient time for extensive RSSI averaging (50–100 samples per position)
- Static nodes eliminate motion-induced RSSI fluctuations
- Controlled indoor environment with minimal external interference

5.5.6 Phase 2: Multi-Target (Sequential Broadcasts) Mode

Expected Precision: ± 10 –15 cm

Phase 2 introduces multiple target nodes transmitting sequentially in time slots. The slight degradation in accuracy compared to Phase 1 is due to reduced sampling time per target and potential timing synchronization errors.

Performance Characteristics:

- **Central Region:** ± 10 cm precision with proper time-slot management
- **Typical Performance:** ± 12 –13 cm across the testbed
- **Edge Regions:** ± 15 –18 cm due to combined effects of geometry and reduced sampling

Factors Affecting Accuracy:

- **Time-Slot Duration:** Shorter slots (100 ms) reduce samples per position, increasing RSSI variance
- **Synchronization Errors:** Clock drift between nodes can cause slot overlap, leading to occasional packet collisions
- **Sequential Latency:** Position updates for each target occur at intervals (e.g., every 300 ms for 3 targets), reducing temporal resolution

5.5.7 Phase 3: All-Nodes / Mesh Mode

Expected Precision: ± 20 cm (relative positioning)

Phase 3 operates in a fully distributed mesh configuration where all nodes act as both transmitters and receivers [9, 10]. The positioning accuracy is specified as "relative" because the system estimates inter-node distances and relative positions rather than absolute coordinates.

Performance Characteristics:

- **Pairwise Distance Estimation:** ± 15 – 20 cm accuracy for direct node-to-node distance measurements
- **Network-Wide Positioning:** ± 20 – 25 cm for positions derived from multilateration across the entire mesh
- **Anchor-Free Operation:** Accuracy degrades without fixed reference anchors, as errors accumulate through the network

Factors Affecting Accuracy:

- **Increased RF Congestion:** All nodes transmit periodically, elevating the noise floor and increasing collision probability
- **Bidirectional RSSI Asymmetry:** RSSI from node A to B may differ from B to A due to antenna patterns and local obstacles
- **Error Propagation:** In mesh networks, distance estimation errors propagate through the graph, compounding positioning uncertainty
- **Computational Complexity:** Graph optimization algorithms (MDS, SLAM) introduce additional approximation errors [18, 19]

5.5.8 Results Summary Table

Table 5.4 summarizes the expected positioning performance across all three system phases.

Table 5.4: Expected Positioning Accuracy by System Phase

System Phase	Expected Precision	Update Rate	Scalability
Phase 1: Single Target	± 5 – 10 cm	10 Hz	1 target
Phase 2: Multi-Target	± 10 – 15 cm	3–5 Hz per target	3–5 targets
Phase 3: Mesh Mode	± 20 cm (relative)	1–2 Hz	6–10 nodes

5.5.9 Factors Affecting Accuracy

Several environmental and system factors influence positioning accuracy across all phases:

Environmental Factors

- **Multipath Fading:** Reflections from walls, furniture, and metallic objects create multiple signal paths, causing RSSI fluctuations of ± 5 –10 dBm.
- **Interference:** Wi-Fi networks, Bluetooth devices, and microwave ovens operating in the 2.4 GHz band can degrade RSSI measurements.
- **Temperature and Humidity:** RF propagation characteristics vary with environmental conditions, affecting path-loss parameters.
- **Human Presence:** People moving near the testbed absorb and reflect RF signals, introducing dynamic RSSI variations.

System Factors

- **Anchor Geometry:** Poor anchor placement (e.g., collinear anchors) increases Geometric Dilution of Precision (GDOP), amplifying distance estimation errors.
- **Calibration Quality:** Inaccurate path-loss parameters (A , n) directly translate to distance estimation errors.
- **Antenna Orientation:** ESP32 PCB antennas have directional patterns; misalignment can cause ± 3 –5 dBm RSSI variations.
- **Sampling Rate:** Insufficient RSSI samples per position increase variance in distance estimates.

Mitigation Strategies

- **Median Filtering:** Reduces impact of outliers and interference spikes
- **Periodic Recalibration:** Adapts to changing environmental conditions
- **Optimal Anchor Placement:** Maximizes triangulation accuracy by ensuring good geometric diversity
- **Increased Anchor Density:** Adding more anchors (5–6 instead of 4) improves overdetermined system robustness

5.5.10 Validation Methodology

The expected results will be validated through systematic experimental testing:

1. **Grid-Based Testing:** Place target at known grid positions and measure positioning error

2. **Statistical Analysis:** Compute mean error, standard deviation, and 95th percentile error across all test positions
3. **Heatmap Visualization:** Generate spatial error distribution maps to identify problematic regions
4. **Comparative Analysis:** Compare trilateration vs. ML regression performance on identical test data

The validation process will confirm whether the system meets the expected precision targets and identify areas for improvement in future iterations.

Chapter 6

Conclusion and Future Work

6.1 Summary of Achievements

This project successfully established a comprehensive framework for a scalable, low-cost Indoor Positioning System based on ESP32 microcontrollers and the ESP-NOW protocol [3, 4]. The work demonstrates that high-precision localization can be achieved using readily available, inexpensive hardware through a carefully designed phased architectural approach [1].

The primary achievements of this project include:

6.1.1 Phased Architecture Implementation

The project introduced a three-phase architectural evolution that ensures compatibility and incremental complexity management. Phase 1 (Hands-Free Single Target) establishes the foundational calibration protocol and validates the path-loss model with expected precision of $\pm 5\text{--}10$ cm. Phase 2 (Multi-Target Sequential) extends the system to handle multiple concurrent targets through time-slot management, maintaining precision within $\pm 10\text{--}15$ cm. Phase 3 (All-Nodes Mesh Network) provides the ultimate goal of fully distributed autonomous positioning with relative precision of ± 20 cm. This phased approach allows each stage to build upon the previous one while maintaining backward compatibility, ensuring that lessons learned and calibration data from earlier phases remain valid.

6.1.2 Calibration Protocol Establishment

A rigorous calibration protocol was developed to derive environment-specific path-loss constants (A and n) from the standard RSSI-distance relationship: $d = 10^{(A - \text{RSSI})/(10n)}$ [5, 13]. The protocol requires collecting 50–100 RSSI samples per anchor at known positions to establish accurate baseline parameters [6]. This calibration phase eliminates environmental variables and provides the foundation for all subsequent positioning operations. The protocol emphasizes static node placement during calibration to minimize motion-induced interference and multipath effects [14].

6.1.3 Machine Learning Model Baseline

An MLPRegressor-based machine learning model was developed and evaluated using synthetic data to explore direct RSSI-to-coordinate mapping as an alternative to traditional trilateration [8, 12]. The model architecture consists of two hidden layers with 64 neurons each, using ReLU activation functions and StandardScaler normalization. Initial results on synthetic data demonstrate the feasibility of this approach, with performance metrics (MSE and R^2 scores) providing a baseline for future improvement with real experimental data [20]. The feasibility analysis identified concrete strategies for enhancement, including increased training data volume, feature engineering techniques, and hyperparameter optimization.

6.1.4 Low-Cost Reproducible Approach

The entire system is designed around commercially available ESP32 development boards, requiring minimal additional hardware (batteries, OLED displays for debugging) [3]. The testbed operates on a compact 70×50 cm grid, making it suitable for laboratory environments and educational settings. The use of ESP-NOW protocol eliminates the need for complex networking infrastructure, reducing both cost and setup complexity [4]. This reproducibility ensures that the system can be easily replicated by other researchers or students for further experimentation and validation [22].

6.2 Key Takeaways

The development and analysis of this Indoor Positioning System revealed several critical operational principles that are essential for successful RSSI-based localization:

6.2.1 Importance of Static Calibration

Accurate localization fundamentally depends on establishing a reliable path-loss model before attempting multi-node operations [5]. The calibration phase must be conducted with all nodes in static positions to eliminate motion-induced RSSI fluctuations [6]. Environmental factors such as multipath fading, temperature variations, and RF interference significantly affect the path-loss constants A and n , making environment-specific calibration mandatory rather than optional [23]. Attempting to skip or shortcut the calibration process inevitably leads to degraded positioning accuracy.

6.2.2 Sequential Timing for Collision Avoidance

In multi-target scenarios, uncoordinated simultaneous broadcasts create signal collisions that corrupt RSSI measurements and make source identification ambiguous [23]. The sequential time-slot protocol, with carefully managed intervals (e.g., 100 ms spacing), is essential for

maintaining data integrity. Each target must broadcast during its designated time slot, with anchors tagging received data with both target ID and timestamp. This approach prevents RF noise floor elevation and ensures that each RSSI measurement can be unambiguously attributed to its source [22].

6.2.3 RSSI Consistency Over Sample Count

Data quality is more important than data quantity for RSSI-based positioning [14]. Collecting 50–100 samples per measurement point and applying filtering techniques (moving average or median filtering) produces more reliable results than using thousands of unfiltered instantaneous measurements [6]. The inherent variability in RSSI due to multipath effects and environmental noise means that statistical aggregation is necessary to extract meaningful distance information [15]. This principle applies across all three phases of the system architecture.

6.2.4 Distributed Mesh Autonomy Benefits

The All-Nodes Mesh Network architecture represents the only viable path toward fully autonomous positioning without centralized infrastructure [9]. By enabling bidirectional pairwise RSSI measurements between all nodes, the system constructs a comprehensive adjacency map that can be processed using graph optimization techniques [10]. This distributed approach eliminates single points of failure, reduces communication bottlenecks, and enables scalability to larger networks. The mesh architecture also supports advanced algorithms such as SLAM-style positioning and multilateration/MDS techniques that leverage the complete network topology [18, 19].

6.3 Evolution Roadmap

The three-phase architecture provides a clear roadmap for system evolution, with each stage building upon the achievements of the previous one:

6.3.1 Stage 1: Single Target (Static Calibration)

This initial stage focuses on establishing the fundamental positioning capability with a single mobile target and multiple fixed anchors [7]. The primary objectives are deriving accurate path-loss constants (A, n) through systematic calibration, validating trilateration algorithms with known ground truth positions, and achieving target precision of $\pm 5\text{--}10\text{ cm}$ [17]. This stage has been completed in the current project, providing the foundation for subsequent phases.

6.3.2 Stage 2: Sequential Multi-Target

The second stage extends the system to handle multiple concurrent targets through time-slot management. Implementation involves integrating the sequential broadcast protocol, validating collision avoidance mechanisms, and testing with 2–4 simultaneous targets. The target precision for this stage is $\pm 10\text{--}15$ cm, with the slight degradation compared to Stage 1 attributed to increased system complexity and potential timing synchronization challenges.

6.3.3 Stage 3: All-Nodes Mesh Network

The final stage implements fully distributed positioning where all nodes act as both targets and anchors. This requires bidirectional ESP-NOW communication, construction of complete RSSI adjacency maps, and application of advanced multilateration or graph optimization algorithms. The target relative precision is ± 20 cm, reflecting the increased complexity of distributed computation and the absence of fixed reference anchors with known absolute positions.

6.3.4 Model Improvement with Real Data

As real experimental data becomes available from the physical testbed, the machine learning model can be significantly enhanced [8]. The synthetic dataset used for baseline evaluation contains only 200 samples, whereas a comprehensive real-world dataset could include thousands of measurements across the entire 70×50 cm grid. Feature engineering techniques such as RSSI ratios between anchor pairs, signal gradient calculations, and temporal consistency metrics can be incorporated to improve model performance [21]. Hyperparameter tuning and exploration of ensemble methods (Random Forest, Gradient Boosting) may further enhance accuracy, potentially surpassing traditional trilateration approaches in complex multipath environments [20].

6.3.5 Advanced Algorithms

Future iterations of the mesh network can incorporate sophisticated positioning algorithms beyond basic trilateration [17]. Graph optimization techniques, similar to those used in Simultaneous Localization and Mapping (SLAM), can process the complete adjacency map to refine position estimates iteratively [10]. Multidimensional Scaling (MDS) algorithms can transform the pairwise distance matrix into a coherent coordinate system [19, 18]. These advanced approaches leverage the rich information available in the mesh topology to achieve better accuracy and robustness than single-point trilateration methods [25].

6.4 Future Work

Several promising directions exist for extending and enhancing the Indoor Positioning System:

6.4.1 Real-World Deployment and Testing

The current system has been designed and validated using synthetic data and theoretical analysis. The next critical step is comprehensive real-world testing on the physical testbed [6]. This involves deploying the hardware across the 70×50 cm grid, conducting extensive calibration measurements, collecting training data for the ML model, and validating positioning accuracy against ground truth measurements. Real-world testing will reveal practical challenges such as battery life management, thermal effects on RSSI stability, and the impact of human presence on signal propagation [23].

6.4.2 Dynamic Environment Adaptation

Current calibration assumes static environmental conditions, but real-world deployments must contend with dynamic factors such as temperature variations, humidity changes, and moving obstacles [16]. Future work should explore adaptive calibration techniques that can detect environmental changes and adjust path-loss parameters in real-time [23]. Machine learning approaches could be employed to model the relationship between environmental sensors (temperature, humidity) and RSSI behavior, enabling predictive compensation for environmental effects [20].

6.4.3 Integration with Existing Infrastructure

The ESP32 platform supports multiple communication protocols including Wi-Fi and Bluetooth in addition to ESP-NOW [3]. Future implementations could integrate the positioning system with existing wireless infrastructure, using Wi-Fi for data logging and remote monitoring while maintaining ESP-NOW for time-critical positioning operations [28]. Integration with cloud services could enable centralized data analysis, remote system management, and multi-site deployment coordination.

6.4.4 Scalability to Larger Networks

While the current testbed uses 4–5 anchor nodes covering a 70×50 cm area, practical applications may require coverage of much larger spaces [2]. Scaling to larger networks introduces challenges in time-slot management (more nodes require longer cycle times), communication range limitations, and computational complexity of graph optimization algorithms [9]. Future research should investigate hierarchical network architectures, where local clusters of nodes perform positioning independently and coordinate through gateway nodes to maintain global consistency [10].

6.4.5 Machine Learning Model Improvements

Beyond the baseline MLPRegressor model, several advanced machine learning techniques warrant investigation [20]. Deep neural networks with additional hidden layers and more sophisticated architectures (e.g., residual connections, attention mechanisms) may capture complex nonlinear relationships between RSSI patterns and positions [8]. Recurrent neural networks (RNNs) or Long Short-Term Memory (LSTM) networks could exploit temporal correlations in sequential RSSI measurements to improve accuracy and reduce sensitivity to instantaneous noise [11, 26]. Transfer learning approaches could enable models trained in one environment to be adapted to new environments with minimal additional calibration data.

6.4.6 Power Optimization

For battery-powered deployments, power consumption becomes a critical constraint. Future work should investigate duty-cycling strategies where nodes alternate between active and sleep modes, wake-on-radio techniques that minimize idle power consumption, and adaptive transmission power control that balances positioning accuracy against energy efficiency. The ESP32's deep sleep capabilities could be leveraged to extend battery life from hours to days or weeks, making long-term deployments practical.

6.4.7 Multi-Floor and 3D Positioning

The current system focuses on 2D positioning within a single plane. Extending to 3D positioning requires additional anchors placed at different heights and modifications to the trilateration algorithms to solve for three coordinates instead of two [17]. Multi-floor positioning in buildings introduces additional challenges due to floor attenuation effects and the need for vertical anchor placement [16]. Future research could explore hybrid approaches combining RSSI-based horizontal positioning with barometric pressure sensors for vertical positioning [28].

6.5 Concluding Remarks

This project demonstrates that accurate, low-cost indoor positioning is achievable using commodity ESP32 hardware and the ESP-NOW protocol [3, 4]. The phased architectural approach provides a practical roadmap from initial calibration through multi-target operation to fully distributed mesh networks [9]. The combination of traditional trilateration techniques with machine learning approaches offers flexibility to optimize for different deployment scenarios and accuracy requirements [8, 17]. The foundation established by this work enables future researchers and developers to build upon these principles and extend the system toward practical real-world applications in asset tracking, robotics navigation, and location-based services [1, 16].

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